

Does Peter Piper Pick Pepper Inattentively? Consumer Inattention to Package Content

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We study consumer inattention to product downsizing, whereby manufacturers raise unit prices by reducing package content. We develop a novel method to recover inattention within a random coefficient discrete choice framework. Applying our method to the pepper industry, where thirty downsizing episodes affected a third of category sales, we find that almost no consumers notice downsizing and that they are more sensitive to package prices than to net weight. Making downsizing salient raises consumer surplus, while banning downsizing generates only half as much consumer surplus gain because restoring package content raises marginal costs and hence prices.

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Product downsizing, or shrinkflation, refers to the practice of reducing package content while maintaining the package price, so that firms effectively raise unit prices. This practice occurs across a wide range of industries. In 2026, Krusteaz reduced the content of its buttermilk pancake mix boxes by 32%, from 30 to 20.5 ounces; Hershey’s decreased its bags of snack size chocolate by 9%, from 19.8 to 18 ounces; and Charmin reduced the sheet count of its Ultra Soft Mega rolls by 7%, from 224 to 208 (Dworsky, 2026). While some manufacturers adjust their packaging to reflect the reduced content, others do not, making detection difficult for consumers.

Detecting downsizing requires consumers to process package content information correctly, yet most firms do not explicitly disclose content reductions. Because many consumers rely on visual estimates rather than on the net weight printed on the label, they may fail to notice reductions in content; for these consumers, downsizing amounts to a hidden unit price increase.

Rising grocery prices in recent years have amplified concerns that downsizing is used to shroud unit price increases, prompting regulatory scrutiny, especially in Europe. Since July 2024, France has required retailers to notify customers when a product is downsized (Adamson, 2024). Austria recently enacted legislation requiring retailers to provide notice when a content decrease results in a per-unit price increase of more than 3% (Niederkofler and Sayahpour, 2026). In the U.S., proposed legislation would direct the Federal Trade Commission to prohibit downsizing as a deceptive practice.¹ The effectiveness and consequences of these measures, in terms of prices and welfare, crucially depend on the extent of consumer inattention.

In this paper, we study whether consumers are inattentive to downsizing and explore the welfare implications of inattention. Using Nielsen scanner data on pepper purchases from 2014 to 2016, we document 30 downsizing events, accounting for approximately 34% of industry sales. The most prominent episode involved McCormick, the industry’s largest firm, which responded to a 2014 spike in wholesale pepper prices by reducing the content of several black pepper products while leaving the external packaging unchanged. The episode triggered several consumer lawsuits, including *In re: McCormick & Co.* (2019). Descriptive evidence suggests widespread inattention: despite substantial increases in the unit prices of downsized products, around four times the increase observed for non-downsized products, their market shares remained stable.

Motivated by these findings, we develop a novel method to recover inattention within a random-coefficient discrete choice framework. In our model, inattentive consumers evaluate product utility on the basis of the product’s original net weight, so that the change in net weight resulting from downsizing enters utility as an additional product characteristic. The random coefficient on this characteristic follows a Bernoulli distribution, where the probability of success corresponds to the likelihood that a consumer is attentive to the change in weight. Estimating the extent of inattention thus amounts to estimating the distribution of a random coefficient, which allows us to rely on standard demand estimation techniques.

Intuitively, our model recovers inattention by comparing how market shares change after downsizing with how they should have changed, given the estimated consumer preferences for net weight. If consumers are inattentive, they do not substitute away from downsized products despite the increase in unit prices, and market shares remain stable; if consumers are attentive, some switch to non-downsized alternatives, and the market share of downsized products declines. The wedge between observed and expected market shares identifies the degree of inattention.

¹Shrinkflation Prevention Act of 2024, S. 3819, 118th Cong. (2024) <https://www.congress.gov/bill/118th-congress/senate-bill/3819/text>

The pepper market is well suited to studying the effects of downsizing. First, pepper is a well-defined product category, with many brands and minimal substitution with other food categories. Second, purchase dynamics are relatively simple: promotions are infrequent, consumers typically buy one unit per purchase, and there is little evidence of stockpiling. Most importantly, the industry features substantial variation in package weight and multiple, well-documented downsizing episodes, allowing us to separately identify consumer preferences for weight, brand loyalty, and inattention. To our knowledge, our study is the first to quantify inattention to downsizing and to disentangle its role from other determinants of consumer inertia, such as brand loyalty.

Our estimates show that nearly all consumers fail to notice downsizing, leading to limited substitution away from downsized products despite their preference for larger packages. In addition, we estimate consumers to be considerably more sensitive to package prices than to net weight.

Motivated by the newly enacted and proposed legislation, we consider two counterfactual scenarios. In the first, we quantify the effect of eliminating consumer inattention, simulating the impact of transparency measures that mandate downsizing disclosures, in the spirit of the 2024 French policy and the 2026 Austrian “Anti-Deceptive Packaging Act”. In markets containing a downsized product, the share of those products falls from 31.2 to 24.1% of pepper purchases, a decline of 23%, while their prices fall only slightly, by less than a cent. Consumer surplus rises by 0.78% of per-capita expenditure on pepper, or roughly \$613,000 a year across the retailers we observe. A back-of-the-envelope extrapolation suggests that, if this per-dollar loss carried over to the rest of the grocery basket, the annual welfare loss from inattention to downsizing would be on the order of half a billion dollars.²

In the second counterfactual, we simulate a complete ban on downsizing. A ban eliminates inattention as well, because the original versions replace the downsized ones; at the same time, restoring the original net weights raises marginal costs, which firms pass through onto final prices almost fully. Under our preferred cost specification, the affected products become \$0.36 more expensive on average, an increase of 9.6% in their own price, averaged across products, and their share falls further, from 31.2 to 22.8%, a decline of 27%. Consumer surplus rises by 0.43%, or around \$327,000 a year, slightly over half of the gain from disclosure. The difference reflects the estimated preferences: providing an additional ounce costs roughly \$0.48, while consumers value it at \$0.36, so the restored content is worth less to consumers than it costs to supply. This conclusion is sensitive to how marginal cost varies with package size. Among the specifications that admit curvature, our preferred one implies the smallest cost increase and is therefore the most favorable to a ban. Under a logarithmic specification, restoring the original weights raises marginal cost by \$0.44 rather than \$0.36, consumer surplus falls by 0.04% instead of rising, and 65% of the affected markets are made worse off.

Taken together, our results suggest that transparency measures, which correct inattention directly, dominate a ban on downsizing, which addresses it only indirectly. Disclosure mandates in the spirit of the French and Austrian legislation therefore appear preferable to an outright prohibition, which generates smaller gains and, under plausible specifications, may even reduce consumer surplus.

²Inattention costs consumers 2.3 cents per dollar spent on downsized pepper. Downsized products account for around 2% of packaged-goods sales in Nielsen data covering 2010–2020 (Janssen and Kasinger, 2026), and U.S. food-at-home spending was about \$1.1 trillion in 2025 (U.S. Department of Agriculture, Economic Research Service, 2025). Because most downsizings are smaller than those we study (Lee (2024) reports a median reduction of 11%, against 10 to 25% in pepper), this figure is likely an upper bound.

RELATED LITERATURE

This paper contributes to the growing literature on downsizing, consumer inattention, and firms' strategic pricing decisions. While downsizing is a widespread practice, empirical research on its prevalence and consequences remains limited. Three recent studies document the extent of downsizing in consumer packaged goods markets in the US. Lee (2024) and Janssen and Kasinger (2026) analyze a decade of retail scanner data and find evidence of downsizing across a wide range of product categories, from candy to pet care. Similarly, Rojas, Jaenicke and Page (2026) document a trend toward smaller package sizes over the past decade mainly driven by product introductions rather than downsizing. Both studies emphasize the role of manufacturers and retailers in driving smaller packages, with manufacturers actively shrinking product sizes and retailers often opting to stock smaller versions even when larger packages are available.

A growing strand of literature examines consumer responses to downsizing in specific industries. Cakir and Balagtas (2014) and Yonezawa and Richards (2016) find that consumers in the ice cream and cereal industries, respectively, are more sensitive to package prices than net weight reductions, implying a muted response to downsizing. Neither study explores why consumers appear to undervalue package size nor does either directly estimate the degree of consumers' attention. Kim (2024) finds that consumers prefer downsizing to price increases and that this preference persists over time; he argues that the benefit-price ratio of a product is higher for downsizing relative to a price increase for fully rational consumers. In contrast, Wan, Harris-Lagoudakis and Crespi (2026) show that consumers in states requiring unit price disclosures are more responsive to downsizing, suggesting that providing clearer size information influences consumer behavior.

More broadly, Janssen and Kasinger (2026) find that, across industries, consumers react more strongly to package price increases than to package size reductions. This asymmetry is absent when firms increase package sizes, suggesting that firms deliberately obscure size reductions while highlighting size increases. These findings align with broader research on consumer inattention, which documents systematic biases in how individuals process product attributes. Consumers frequently ignore explicit size information and instead rely on visual cues when evaluating package volume (Lennard et al., 2001). Visual estimates can be highly inaccurate due to cognitive biases. For instance, consumers perceive tall, narrow packages as larger than short, wide ones of the same volume (Krishna, 2006), and such misperceptions intensify when multiple packaging dimensions change simultaneously (Chandon and Ordabayeva, 2009). In some cases, consumers fail to notice package size reductions of up to 24% (Ordabayeva and Chandon, 2013). We contribute to this literature by exploring whether inattention explains consumers' weak responses to downsizing.

Beyond visual biases, inattention to unit prices also plays a role. Many consumers do not compare unit prices across different sizes of the same brand and often pay a surcharge for larger quantities (Clerides and Courty, 2017). Consumers who fail to compare unit prices within a brand are even less likely to do so across brands, reinforcing the effectiveness of downsizing as a strategy to obscure unit price increases. Measuring the degree of inattention is critical as it determines the extent to which firms can profitably engage in downsizing.

Consumers exhibit inattention and cognitive biases in a variety of settings. A large literature demonstrates that consumers do not pay close attention to shrouded attributes, such as shipping costs (Brown, Hossain and Morgan, 2010) or sales taxes (Chetty, Looney and Kroft, 2009). Consumers may also misperceive product attributes. Allcott (2013) and Grigolon, Reynaert and Verboven (2018), for instance, find that consumers misjudge the value of fuel economy when choosing vehicles. In some cases, particular product attributes receive disproportionate attention. For

example, many consumers overemphasize the left-most digit and are willing to pay higher prices for cars with mileage just below 10,000 miles (Lacetera, Pope and Sydnor, 2012). If cognitive biases can influence large, high-stakes purchases, they are even more likely to affect low-stakes, habitual purchases such as consumer packaged goods.

Inattention in our setting reflects the multiple cognitive biases described above. Consumers may face search frictions that limit their ability to compare unit prices, or firms may engage in obfuscation strategies, such as displaying downsized products more prominently to increase search costs. Visual biases may lead consumers to misjudge package volume, while price-quality heuristics could cause them to associate higher prices with better quality. Rational inattention may also play a role, as consumers, weighing minor cost savings against time constraints, may choose not to engage in detailed unit price comparisons. Importantly, our estimation strategy separates inattention from consumer inertia stemming from brand loyalty by absorbing the latter through a McCormick indicator. McCormick is the dominant firm, accounting for around 42% of dollar sales, and the only name brand to downsize. We also verify robustness to richer controls: controlling directly for persistent product-level demand and allowing McCormick taste to vary across consumers with income leave the inattention estimate unchanged.

Even if consumers are inattentive to downsizing at the time of purchase, firms face potential risks in exploiting this inattention. Consumers may feel deceived and react negatively upon discovering size reductions. Experimental and survey evidence suggests that consumers express a lower willingness to buy from brands that engage in downsizing (Kachersky, 2011; Wilkins, Beckenuyte and Butt, 2016; Evangelidis, 2024). Evangelidis (2024) finds that participants are more likely to perceive downsizing as unfair compared to an equivalent price increase, mainly because they view downsizing as deceptive. In our data, interpurchase times are long relative to the sample period, and we find no evidence of declining demand for downsized brands, suggesting that consumer backlash is limited over the time horizon we study.

Several studies consider firms' decisions to use downsizing from a strategic perspective. Cakir (2022) finds that firms employing downsizing achieve higher pass-through rates, suggesting that downsizing can be an effective profit-enhancing strategy. In contrast, Yonezawa and Richards (2016) argue that price and size are strategic complements, meaning that downsizing intensifies price competition and can reduce a firm's profits. The decision to downsize thus depends not only on consumer inattention but also on market structure and competitive dynamics. Our framework accounts for strategic pricing decisions: marginal costs are recovered from a multiproduct Bertrand–Nash equilibrium under the observed ownership structure, so markups adjust across a firm's entire line and prices are adjusted in our counterfactuals.

Finally, this paper contributes to the broader literature on modeling consumer inattention. Several studies have developed methods to detect and quantify inattention to product attributes. Abaluck and Compiani (2025) propose a test based on cross-derivatives of choice probabilities. Their method assumes that consumers fully ignore the hidden attribute when searching, making it unsuitable for our setting. Brown and Jeon (2024) provide a method for recovering consumer information-processing strategies grounded in a rational inattention framework. Their approach relies on restrictions on the prior distribution of product utilities and, consequently, the information processing strategy. In contrast, our approach estimates inattention without imposing restrictions on the prior distribution of product utilities or on the information-processing strategy, though it does not explicitly model the process by which consumers become inattentive.

Our model also connects with the literature on random coefficient discrete choice models with

discrete consumer types (e.g. Doi, 2022; Greene and Hensher, 2013; Fox et al., 2011; Berry and Jia, 2010; Berry, Carnall and Spiller, 2006). A key distinction is that, in our setting, the discrete types correspond to attention and inattention. Our approach is most closely related to Greene and Hensher (2013), incorporating both discrete and continuously distributed random variables. However, unlike Greene and Hensher (2013), we estimate an aggregate demand model following the framework of Berry, Levinsohn and Pakes (1995) rather than relying on maximum likelihood or the Expectation-Maximization algorithm.

I. Data

A. Summary

To analyze downsizing, we use the NielsenIQ Retail Scanner data and the NielsenIQ Consumer Panel data from the Kilts Center at the University of Chicago. The Retail Scanner data provides weekly point-of-sale records from approximately 35,000 stores in the United States, covering over 4 million consumer packaged goods. The Consumer Panel data tracks purchases at the household level, following between 40,000 and 60,000 households. We use the Scanner data from 2014 to 2016 for structural estimation and incorporate individual-level purchase data mainly to inform our modeling.

Pepper products encompass various spices used to add heat and flavor to food. These range from true pepper, *Piper nigrum*, to botanically unrelated chilies such as cayenne and seasoning blends where pepper is the primary ingredient, such as lemon pepper. The scanner data includes 1,468 unique pepper products, which we classify into 22 different varieties. Black and white pepper are the two most popular varieties, accounting for 63.8% of sales. Other major categories include red pepper, cayenne, and black pepper blended with citrus. Many of the remaining categories are niche, with only a few products, and we group them into a single “other” category.

Pepper is a staple seasoning, with most households purchasing it at least once between 2012 and 2016.³ However, despite its widespread use, pepper purchases are infrequent. In any given year, only about 42% of households buy pepper, and households in the micro panel make 0.7 pepper purchases per year on average. Among households purchasing more than once, the median gap between consecutive purchases is 22 weeks, and more than a fifth of gaps exceed a year. The long interpurchase intervals reflect pepper’s high storability. Because pepper is derived from dried berries or chilies, it does not spoil easily but instead loses pungency over time (Feucht, 2019).

While different types of pepper vary in flavor and heat intensity, products within the same variety are highly similar, as they originate from the same plant species. Differences within a variety arise primarily from soil conditions, climate, and processing methods, leading to subtle variations in quality and taste. However, for a given type of pepper, the most prominent differences are in branding and packaging. Between 2014 and 2016, the scanner data includes 241 distinct brands.

Given the high degree of product similarity, many consumers opt for lower-cost store brands, which account for approximately 40% of the market during this period. Most national brands are small and regional, typically holding less than 0.1% of the market. McCormick stands out with approximately 42% of dollar sales. Its parent company, McCormick & Co., dominates the industry, owning three of the top five best-selling brands: McCormick, 5th Season, and Spice

³Authors’ calculation based on a balanced panel from the NielsenIQ Consumer Panel data.

Classics. Collectively, McCormick & Co. controls around 50% of the market. The second-largest firm, B&G Foods, which produces Tone’s and Durkee, holds approximately 2.5% of the market, while few other brands exceed a 2% share.

In addition to the large number of brands, the industry features a wide range of package sizes. In the consumer panel data, nearly all purchases range from 0.4-ounce bags to 32-ounce containers, with substantial variation in between (Figure A1).⁴ Examining the distribution of purchased sizes from 2014 to 2016, the most frequently bought packages are two and four ounces, which correspond to the standard sizes of small and medium tins of black pepper.

Most stores in the scanner data stock these two black pepper sizes, along with many others. Stores carry 14.7 distinct sizes of pepper products in the average month (Figure A2), of which 8.5 are black or white pepper. Assortment breadth varies considerably, partly reflecting differences between the food and mass channels: half of the stores offer eleven or fewer sizes, while the broadest offer more than fifty. Differences in the available sizes across retailers force consumers to substitute toward similarly sized products and provide variation in choice sets that helps separate size preferences from inattention.

In addition to scanner data, we use monthly spot price data for black and white pepper from the International Pepper Community to instrument for price.

B. Estimation Sample

We define a product as a unique UPC in the Nielsen data. Due to the computational demands of structural estimation, we aggregate the data from the store level to the retailer and designated market area (DMA) level and from weekly to monthly frequency.⁵ Specifically, we aggregate the Nielsen scanner data from its original UPC-week-store level to a UPC-retailer-DMA-month level. We sum quantities across weeks within a month and compute a quantity-weighted average monthly price for each product.

As shown by DellaVigna and Gentzkow (2019) and Hitsch, Hortaçsu and Lin (2021), prices and product assortments are highly similar within the same retailer, justifying aggregation at the retailer-DMA level, which preserves geographic variation while reducing dimensionality. Temporal aggregation is also appropriate, given the infrequency of pepper purchases and the absence of evidence of sales-driven stockpiling. The likelihood of consumers purchasing multiple units within a given month is low, ensuring that aggregation does not distort demand patterns.

We restrict the estimation sample to food stores and mass merchandisers. Drug stores are excluded because of their limited pepper assortments. When a store carries very few varieties, demand variation is driven primarily by assortment constraints rather than consumer preferences, which can bias estimates of price and size coefficients. For the same reason, we impose the additional requirement that the retailer offers at least seven distinct sizes in a given month. This restriction ensures that the estimation sample consists of retailers where consumers face a sufficiently rich choice set for the observed substitution patterns to be informative about their underlying preferences.⁶

⁴There are a few outliers, including sales of 80-ounce containers.

⁵Following DellaVigna and Gentzkow (2019) and others, we define a retailer as a combination of the fields *parent_code* and *retailer_code*.

⁶The choice of seven sizes balances two considerations: it is high enough to ensure a meaningful choice set, but low enough to avoid over-selecting the sample. We estimated the model under alternative cutoffs and find that the results are robust to this threshold. Higher cutoffs yield similar estimates but progressively reduce the sample size, particularly among mass merchandisers.

In the final estimation sample, the typical retailer offers around 24 distinct sizes of pepper products, with retailers carrying between 7 (the cutoff) and 73 sizes. Figure A3 presents the distribution of unique sizes offered by retailers in a given month. As in the store-level data, some retailers offer a more limited selection. However, while aggregation increases the number of distinct sizes observed, the overall variation remains comparable to that in the store-level data.

The final estimation sample consists of 1,217 retailer-DMA combinations, yielding 1,355,078 product-market observations. For simplicity, we refer to each retailer-DMA combination as a retailer.

During the sample period, the mean price of a pepper product was \$3.96, with most prices ranging from \$0.99 to \$7.99. This corresponds to an average unit price of \$1.85 per ounce, with the vast majority of unit prices falling between \$0.45 and \$4.00.

II. Downsizing in the Pepper Industry

A. Background

Downsizing in the pepper market came in response to rising commodity costs. From 2009 to 2014, wholesale black pepper prices were rising due to growing demand in emerging markets (Figure A4). With prices trending upward, a poor harvest in 2014 caused the wholesale prices to spike. Over the course of 2014, the wholesale price of black pepper increased by over 30%. Manufacturers responded to this sudden cost increase in different ways. Most increased their product prices, while others, including McCormick and Spice Classics, reduced the content of select black pepper products. Except for one blend and one red pepper product, these firms only adjusted their black pepper products.

Faced with higher wholesale prices, retailers responded by adjusting their product offerings, with some phasing out larger products for smaller ones. McCormick offered the retailers whose private-label pepper it manufactures a choice between accepting reduced fill levels and paying more to keep the original weights; most accepted the smaller sizes, while several private-label brands declined and took a wholesale price increase instead (In re: McCormick & Co., 2019, p. 13). McCormick states that it wanted to preserve product quality while avoiding large price increases (Hughlett, 2015). Several consumer lawsuits, however, allege that McCormick sought to exploit consumer inattention, citing an internal email in which a McCormick employee writes, “*keep in mind consumers do not really know the fill level right now*” (In re: McCormick & Co., 2019, pp. 13–14, quoting Fegan Decl. ¶181). Regardless of the reason for downsizing, the strategy’s effectiveness depends on the degree of consumer inattention.

McCormick downsized its black pepper products and store brands in February 2015. McCormick initially downsized its products by reducing the fill levels while keeping the packaging unchanged, so that the original and downsized tins were indistinguishable on the shelf except for the net weight printed on the label.⁷ The company eventually adjusted its package to reflect the reduced content in the middle of 2016 (In re: McCormick & Co., 2019).

⁷Photographs of the original and downsized tins side by side on a store shelf, and of the original and downsized grinders with the labels removed, are reproduced in the complaint in *Watkins v. McCormick* (2015, p. 10) and in the class-certification opinion in *In re: McCormick & Co.* (2019, pp. 14–15), available at https://www.govinfo.gov/content/pkg/USCOURTS-dcd-1_15-mc-01825/pdf/USCOURTS-dcd-1_15-mc-01825-6.pdf.

B. Identification of Downsized Products

To identify which products in the data are downsized versions of existing products, we compare time series of unit sales for candidate product pairs that share the same brand, pepper type, and packaging but differ in net weight. We look for the characteristic pattern of declining sales for the original product accompanied by increasing sales of a slightly smaller replacement. We validate our identifications against the court filings in *In re: McCormick & Co.* (2019), which documents the specific products and timing of McCormick’s downsizing decisions. Appendix A.A1 describes the procedure in detail.

From this procedure, we identified 30 downsizing episodes: 15 name-brand and 15 private-label downsizing episodes. Table A2 provides a complete list of the name-brand products.⁸ Figure A5 shows the distribution of the resulting downsizing magnitudes, measured both as a percentage of the original net weight and in absolute ounces. Downsizing is rare, and the reductions cluster at a small number of focal magnitudes: in proportional terms most fall between 10 and 25%, with a mode at roughly one-quarter of the original size, while in absolute terms they concentrate around one and two ounces. This clustering at round magnitudes, with gaps in between, is consistent with coarse, discrete size adjustments on a limited set of package formats rather than fine, product-specific calibration.

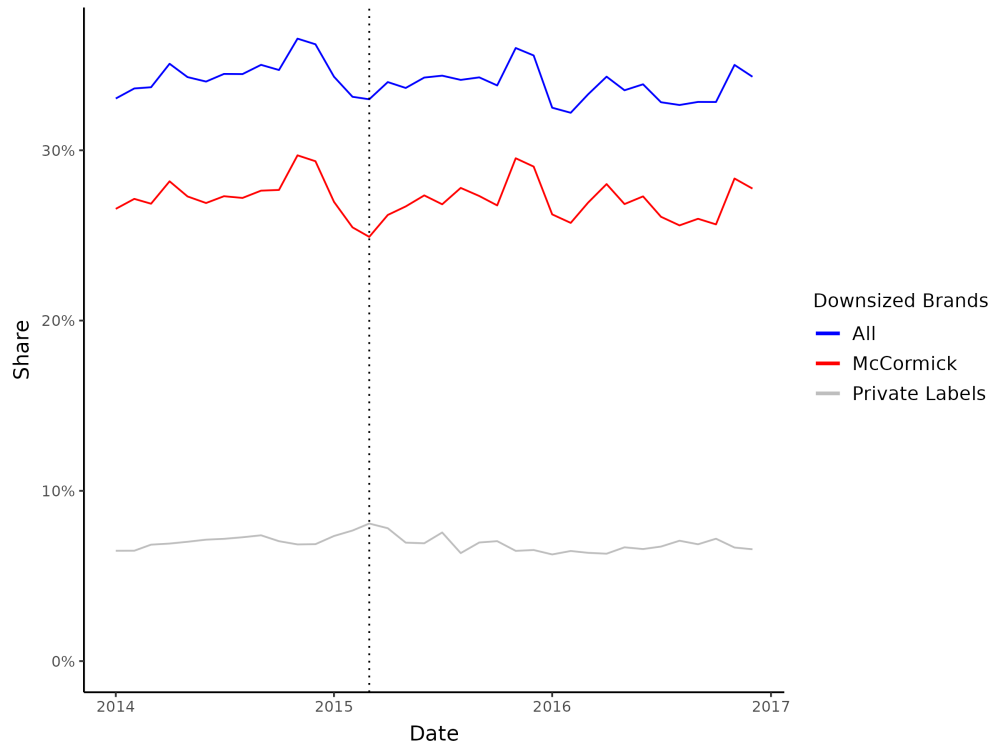
C. Market Shares and Prices

While the number of downsized products is small, these products account for around 34.1% of all pepper sales in the data. McCormick’s downsized products account for around 27.2% of sales and the downsized private-label products for around 6.9%.

Figure 1 shows the share of the downsized products, separately by McCormick and private labels. The dashed line represents the date when McCormick started to ship its downsized products. The shares include the sales of both the original and downsized versions. Before downsizing, the share represents the original product. Just after downsizing, the shares represent a combination of downsized and original versions as retailers sell out their existing inventory of the original version and replace it with the new version. By the middle of 2016, the share mainly reflects the downsized version.

⁸Spice Classics downsized two products, but their sales are negligible; we therefore focus on downsizing by McCormick and private labels.

Figure 1. : Market shares of the downsized products



As the figure shows, the share of the downsized products is stable over time both for McCormick and the private-label products. The stability of the shares is surprising given the unit price changes. Figure 2 shows the average price per ounce of the downsized products relative to the non-downsized products. Comparing 2014 and 2016 averages, the average unit price of the non-downsized products increased by \$0.09. In contrast, the average unit price of the downsized product increased by around \$0.38, more than four times the increase for non-downsized products. Despite this large increase in relative unit prices, there was not a corresponding decline in the share of the downsized products.

Figure 2. : Unit prices



Several possible reasons can explain the observed trends in shares and unit prices. The first is brand loyalty. Consumers may have a strong attachment to the downsized brand and as a result, do not substitute away from these products despite the increase in unit prices. However, strong brand attachment is difficult to rationalize given the homogeneity of the product category. Moreover, our structural model absorbs McCormick's brand premium through a McCormick indicator, so the inattention we estimate is net of brand loyalty.

Another possibility is that fully rational consumers are more sensitive to package prices than to package content. Kim (2024) argues that consumers may not respond to content reductions because, under certain demand conditions, downsizing generates more consumer surplus than an equivalent price increase. Our structural model accounts for this possibility by estimating consumers' sensitivity to net weight, including heterogeneous responses through a random coefficient.

We focus on a third possibility: that consumers are inattentive. Consumers do not respond to downsizing because they do not notice it, failing to use the available information on net weight and unit prices to their detriment.

These explanations are not mutually exclusive. Consumers may be, at the same time, inattentive to downsizing and only weakly responsive to the reduction in net weight. The two effects play distinct roles: inattention determines whether consumers notice the change, while size sensitivity determines how much they substitute if they do. Where size sensitivity is low, even fully attentive

consumers would move away from the downsized products only modestly. Our model disentangle the two effects separately.

III. Evidence of Inattention

Sales at stores that sell both the original and downsized versions at the same time provide evidence of inattention. During the transition period from the old product to the new one, some stores start to stock the downsized version before selling out of the original, resulting in both versions appearing together and, in some cases, side by side.⁹

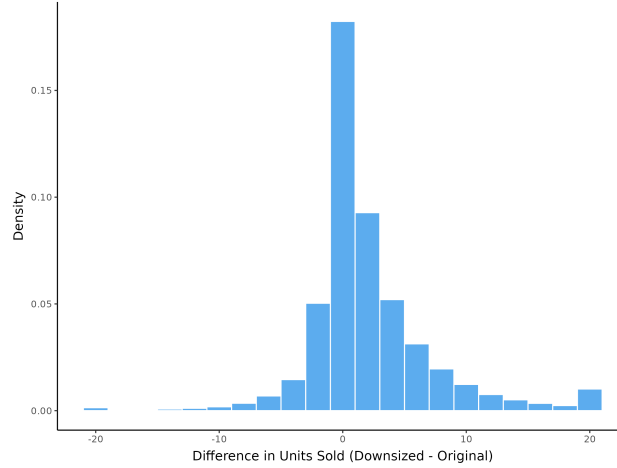
In the data, we observe 159,878 store-product-weeks where consumers buy both the original and downsized version of a product in the same week. This does not mean that the products necessarily appear together on store shelves. Because stores report units sold weekly, a store could sell out of the original product on Wednesday and start selling the new product on Thursday. In that case, both products would appear in the scanner data in the same week without having been available simultaneously. To minimize this possibility, we remove the week in which the downsized version first appears and the week in which the original last sells, keeping only store-product-weeks strictly between them in which both products record positive sales; those boundary weeks are the ones in which a store is most likely to have switched from one version to the other mid-week. This leaves 107,752 store-product-weeks in the overlap sample.

Figure 3 shows a histogram of the difference in sales between the downsized and original products for pairs that have the same package price. The plot shows that the downsized product is more popular than the original: consumers are choosing to pay a higher unit price for less pepper. This pattern cannot be plausibly attributed to brand loyalty or habit, as the two versions share the same brand, format, and price. Given that the downsized product differs primarily in net weight, it is hard to explain this difference without appealing to inattention.

One concern is that the higher sales of the downsized product may partly reflect shelf availability, as stores transitioning to the new product likely stock more units of the downsized version. While we cannot observe shelf quantities directly, stocking asymmetries during the transition could favor the downsized version. Attentive consumers, however, would have no reason to prefer it, as the original offers strictly more content at the same price. We therefore read this pattern as suggestive; the estimates on the non-overlap sample (Section VI.B), which do not rely on the transition period, provide independent evidence.

⁹For an example, see page 10 of the complaint in *Watkins v. McCormick* (2015).

Figure 3. : Histogram of sales differences for downsized product pairs with equal package prices



Notes: The difference in units is top/bottom coded at the values -20 and 20 .

To extend this analysis to product pairs with different packaging prices, we regress the difference in the number of units sold of the downsized product and the original product on the difference in the package prices. If consumers are fully attentive, they should switch quickly to the original product as the downsized version becomes relatively pricier. Table 1 reports those results. To mitigate the impact of outliers on our regressions, we winsorize the top and bottom 1% of both the difference in units sold and the difference in prices. Winsorized results are reported in specifications (3) and (4).

In many weeks only one of the two products records a sale, even when both are available on store shelves, so the price of the unsold version is unobserved. We impute the missing price and unit-sales differences using the most recently observed values for that store-product pair, looking back up to 26 weeks (last observation carried forward). Imputation increases the sample from 107,752 to 398,102 store-product-weeks, and the resulting estimates, in specifications (5) and (6), are close to those in specifications (1) and (2), which require both products to record positive sales and are therefore unaffected by stockouts.

Table 1—: Difference in sold units between downsized products and their original counterparts

	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	2.320 (0.026)		2.400 (0.019)		2.772 (0.012)	
Package Price Difference	-0.490 (0.053)	-0.462 (0.056)	-0.787 (0.047)	-0.783 (0.049)	-0.441 (0.028)	-0.412 (0.029)
Num. Obs.	107,752	107,752	107,752	107,752	398,102	398,102
Product & Time FE	NO	YES	NO	YES	NO	YES
Winsorize	NO	NO	YES	YES	NO	NO
Imputed Prices	NO	NO	NO	NO	YES	YES

The positive intercept indicates that, even at equal package prices, consumers purchase 2.3 to 2.8 more units of the downsized product per store-product-week, consistent with the histogram evidence. The negative coefficient on the package price difference implies that consumers do switch to the original product as the downsized version becomes more expensive. However, the magnitude is small: even with the largest estimate of -0.787 in specification (3), the downsized product would need to be approximately \$3 more expensive before sales equalize. For context, the typical pepper product in our sample retails for under \$5, so the required price difference is comparable to the price of the product itself. Consumers are not responsive enough to price differences, consistent with inattention to package content.¹⁰

Overall, these descriptive results suggest that consumers fail to notice downsizing. Under equal prices, they choose the downsized product more often than the original version. Generally, they do not respond nearly enough to differences in the package price, preferring instead to buy the more expensive downsized product. We next model this inattention directly.

IV. Product Choice under Inattention

A. Model

In period t , M_{kt} consumers visit retailer-market k to purchase pepper.¹¹ Consumers choose one product from the set of inside goods J_{kt} or the outside option 0. Consumer i 's *actual* utility from purchasing product j is:

$$(1) \quad U_{ijkt}^a = x_{jkt}\beta + \gamma_i z_{jkt} - \alpha_i p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt},$$

where x_{jkt} is a set of observable characteristics; p_{jkt} is the price; z_{jkt} is the current net weight; ξ_{jkt} is an unobserved (to the econometrician) product attribute; and ε_{ijkt} is a random shock.¹² The utility of the outside option is:

$$(2) \quad U_{i0kt}^a = 0 + \varepsilon_{i0kt}.$$

Some consumers may be inattentive and fail to notice changes in net weight. They may remember the old weight and assume that weight has not changed since their last purchase. Other consumers may judge product weights by package dimensions and mistakenly conclude that the downsized product weighs the same as rivals with identical packaging. Regardless of the source of inattention, inattentive consumers evaluate the product using its original product weight, whereas attentive consumers evaluate the product using its current product weight.¹³

¹⁰When we replace the package price difference with the unit price difference, the coefficients remain negative (-0.797 without fixed effects, -2.008 with imputation and fixed effects) and the qualitative pattern is unchanged. The magnitudes are not directly comparable to those in Table 1.

¹¹Throughout, k indexes a retailer-DMA market.

¹²We write the price term as $-\alpha_i p_{jkt}$ with $\alpha_i > 0$, so the coefficient on price in utility is $-\alpha_i$. All tables report this utility coefficient and it is therefore negative. Because ν_i^p is symmetric about zero, the sign of σ^p is not separately identified, so writing $+\sigma^p \nu_i^p p_{jkt}$ in μ_{ijkt} is equivalent to writing $-\sigma^p \nu_i^p p_{jkt}$.

¹³We implicitly assume that consumers evaluate a downsized product at its original net weight, either because they recall the weight from prior purchases or because they judge content by the external package dimensions, which remained unchanged at least until mid-2016 (In re: McCormick & Co., 2019). The two mechanisms differ for consumers who enter the category after downsizing. Under the memory mechanism, an entrant has no remembered weight and evaluates the product at its current weight, behaving exactly as an attentive consumer would; the model cannot distinguish the two, so the share of entrants is a lower bound on the attention probability we estimate. Our estimate of η is essentially zero, which requires entrants to be a

Consumers are either attentive or inattentive to downsizing. An *inattentive* consumer evaluates the downsized product j using its original weight and *perceives* their utility from j as:

$$\begin{aligned}
 (3) \quad U_{ijkt}^p &= x_{jkt}\beta + \gamma_i z_{jk0} - \alpha_i p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} \\
 &= x_{jkt}\beta + \gamma_i z_{jkt} + \gamma_i (z_{jk0} - z_{jkt}) - \alpha_i p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} \\
 &= U_{ijkt}^a + \gamma_i \Delta_0 z_{jkt}
 \end{aligned}$$

where z_{jk0} is the original weight before downsizing and $\Delta_0 z_{jkt} = z_{jk0} - z_{jkt}$ is the change in the product weight.¹⁴ In our context, product weights change once or not at all. If the size changes in period t' , $z_{jkt} = z_{jk0}$ for all periods $t < t'$. In contrast to actual utility, perceived utility depends both on the current and original weight. Inattention drives a wedge between the perceived and actual utility for downsized product j equal to $\gamma_i \Delta_0 z_{jkt}$.

For attentive consumers, actual and perceived utility are the same. Let τ_i be an indicator for whether consumer i is attentive. We can write the perceived utility of any consumer as:

$$(4) \quad U_{ijkt}^p = U_{ijkt}^a + (1 - \tau_i) \gamma_i \Delta_0 z_{jkt}$$

where τ_i is an indicator for whether the consumer is attentive. Because types are idiosyncratic and not observable, we can view τ_i as a random coefficient that follows a Bernoulli distribution, where the probability of success η represents the probability of being attentive. In essence, inattention causes the weight change to enter perceived utility as an additional product characteristic with a random coefficient. We denote the cumulative distribution function of the random coefficients as $G(\tau_i, \gamma_i, \alpha_i)$ where $\tau_i | \gamma_i, \alpha_i \sim \text{Bernoulli}(\eta)$.

Assuming that the random taste shock ε is drawn i.i.d. from a Type I extreme value distribution, consumer i 's share for product j at retailer k in period t conditional on the random coefficients is:

$$(5) \quad s_{ijkt}(\beta, \alpha_i, \tau_i, \gamma_i) = \frac{\exp \{x_{jkt}\beta - \alpha_i p_{jkt} + \xi_{jkt} + \gamma_i z_{jkt} + (1 - \tau_i) \gamma_i \Delta_0 z_{jkt}\}}{1 + \sum_{l \in J_{kt}} \exp \{x_{lkt}\beta - \alpha_i p_{lkt} + \xi_{lkt} + \gamma_i z_{lkt} + (1 - \tau_i) \gamma_i \Delta_0 z_{lkt}\}}$$

The model ignores retailer choice. This abstraction is reasonable as consumers select a retailer based on a basket of products rather than just pepper (Thomassen et al., 2017). In the consumer panel data, every household purchases pepper as part of a larger shopping trip and, as such, pepper prices are likely not an important determinant of retailer choice.

B. Generalization

The two-type model can be generalized by allowing consumers to be attentive to some downsized products but not others. With L downsized products, each consumer's type is a vector $\tau_i \in \{0, 1\}^L$, yielding 2^L possible latent classes. The joint distribution $G(\tau, \gamma, \alpha)$ governs the probability of each type. Different assumptions about inattention place different restrictions on the latent

negligible share of buyers. This is plausible in our setting: 80% of households buy pepper at least once between 2012 and 2016, and the median gap between consecutive purchases is 22 weeks, so buyers during the downsizing period are overwhelmingly repeat buyers. Under the visual mechanism the issue does not arise, since entrants misjudge content from the package dimensions in the same way as incumbent buyers.

¹⁴A downsized product receives a new UPC; we link each original and its downsized version into a product family (Appendix A.A1). For a downsized UPC, z_{jk0} is the predecessor's net weight; products that were never downsized have $z_{jk0} = z_{jkt}$ and $\Delta_0 z_{jkt} = 0$. During the transition, the original and downsized versions can coexist in the same choice set.

class structure. For instance, under threshold perception (e.g., Han, Gupta and Lehmann, 2001), consumers who notice small size changes must also notice larger ones, reducing the number of classes from 2^L to $L + 1$. While the modeling framework is flexible enough to accommodate these richer structures, the number of latent classes grows exponentially in the number of products, making estimation infeasible in our setting. We therefore focus on the two-type model; in Section VI.B, we examine robustness to the magnitude of downsizing.

C. Identification

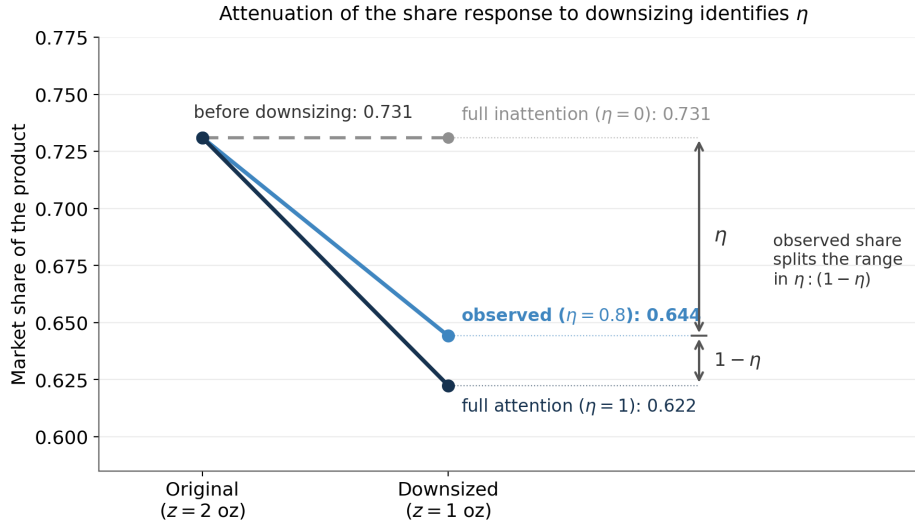
Two sources of variation identify the inattention parameter. The first is retailers that offer the original and downsized versions of the product at the same time. The second is deviations in market shares from consumers' preferences for net weight.

As shown in Section III, many retailers offer both versions of the product at the same time. Differences in the market shares after accounting for other factors point to inattention. However, using the overlap periods to identify inattention warrants caution. Because we aggregate the data to the retailer-DMA-month level, we may incorrectly conclude that the downsized and original products appear side-by-side when in fact they do not: one store may stock only the original while another stocks only the downsized version. Consumers at those stores buy the downsized product without having the option to buy the original, yet the aggregated data record both products in the same choice set. This pattern mimics inattention and would bias the estimated probability of attention downward. As a robustness check, we re-estimate the model both excluding the overlap months entirely and on the overlap months alone, and find the same results (Section VI.B), confirming that our estimate of inattention is not an artifact of the overlap period.

The second source of identification is deviations in relative market shares from consumers' preferences for net weight. When a product's net weight decreases, its share should change in line with consumers' weight preferences when consumers are fully attentive. Inattention will dampen this response. Thus, a smaller than expected change in retailer shares or no change at all would indicate inattention. If consumers are attentive, the difference in the shares of the original and downsized versions will reflect the difference in weight, all else equal. In contrast, if consumers are inattentive, there will be no difference in the shares. The degree of inattention therefore governs how close the observed difference is to the expected difference.

The above logic applies not just to the original product, but to any product, including the outside option. As an illustrative example, consider a retailer that offers a single product with weight z and an outside option. The utility of the product is $u_1 = \gamma z + \varepsilon_1$ with ε drawn from a Type I extreme value distribution. Initially, $z = 2$ and the market share is $s_1 = \frac{e}{1+e} \approx 0.731$, implying $\gamma = \frac{1}{2}$. The product is then downsized to $z = 1$. Under full attention, the share falls to $s_2 = \frac{e^{0.5}}{1+e^{0.5}} \approx 0.622$. Under complete inattention, consumers evaluate the product at the original weight and the share remains at 0.731. With a fraction $\eta = 0.8$ of attentive consumers, the observed share is $s_2 = \eta \frac{e^{0.5}}{1+e^{0.5}} + (1-\eta) \frac{e}{1+e} \approx 0.644$. The degree of attenuation, that is, the distance between the observed and full-attention shares relative to the distance between the full-inattention and full-attention shares, identifies $1 - \eta$. Figure 4 provides an illustration.

Figure 4. : Identification of Inattention: an Illustrative Example



Identification rests on an exclusion restriction on the unobserved quality. A downsized product receives a new UPC, and for that UPC the size change $\Delta_0 z_{jkt}$ is constant across markets and months. We require that its introduction does not shift ξ_{jkt} in a way that co-varies with $\Delta_0 z_{jkt}$, since it would be attributed to inattention. Several such shifts can be dismissed. In the first year post-downsizing, the containers were of the same size as before, with the reduced weight noted in small print, so the downsized product appeared to consumers exactly as its predecessor had (In re: McCormick & Co., 2017); promotions are infrequent in this category, and the products were not otherwise relabeled. The overlap period is informative in this context. During the transition, many retailers stocked both versions at once, in containers of identical size and appearance differing only in the net weight printed on the front. Two such UPCs can hardly differ in unobserved quality. Estimating on the overlap sample alone leaves the attention probability at zero (Section VI.B).

Given this restriction, η is identified by more than a shift in mean utility. Because τ_i is a random coefficient, attentive and inattentive consumers substitute differently across the entire choice set, and the resulting mixture of choice probabilities cannot be absorbed into δ_{jkt} . The Monte Carlo in Appendix B.B2 confirms this: with ξ drawn independently of the size change, our procedure recovers the true attention probability.

A second identification assumption requires that tastes for price and net weight be stable over time: the pre-downsizing period pins down γ , and, under time-invariant tastes, delivers a counterfactual prediction for how shares should respond to downsizing under full attention. If tastes for net weight were changing, post-downsizing share movements could reflect variation in preferences rather than inattention. We assess this assumption using two complementary diagnostics based on a standard logit regression estimated on actual net weights using the Berry (1994) inversion:

$$(6) \quad \log \left(\frac{s_{jkt}}{s_{0kt}} \right) = x_{jkt}\beta + \gamma z_{jkt} - \alpha p_{jkt} + \xi_{jkt},$$

where s_{jkt} denotes the market share of product j at retailer k in period t and s_{0kt} is the outside share. The vector x_{jkt} includes observable product characteristics (a McCormick brand indicator, pepper-type dummies, and an indicator for whole peppercorns). The variable z_{jkt} is net weight in

ounces and p_{jkt} is package price.

First, we assess within-pre-period stability by splitting the pre-downsizing window into two sub-periods and testing whether the price and net-weight coefficients differ across halves. Under time-invariant preferences, these coefficients should be statistically indistinguishable. Second, we compare the net-weight coefficient estimated before downsizing with the same coefficient estimated afterwards. This comparison shows that the attenuation appears only after downsizing, and in the direction inattention predicts.

Appendix B.B1 and Table B1 report the results. We find stable coefficients within the pre-period but a discrete shift coinciding with downsizing.¹⁵

These patterns are consistent with consumer inattention. When inattentive consumers evaluate a downsized product as if it still weighed the original amount, they buy it more often than its actual weight would predict. A regression using actual net weights attributes that unexpectedly high share to insensitivity to weight, biasing the estimated γ downward.

D. Limitations

Our model has two main limitations. First, the model does not microfound the frictions that generate inattention: consumers are either attentive or inattentive, with no intermediate information-acquisition stage. Frameworks based on rational inattention (e.g., Brown and Jeon, 2024) could in principle model information acquisition, but at the cost of specifying prior beliefs over product utility. Two patterns suggest, however, that the inattention we estimate is not primarily a rational response to the cost of acquiring size information. We estimate the same level of inattention in states that mandate unit-price disclosure and in those that do not (Section VI.B), so inattention does not appear to mainly operate through the salience of per-unit prices. Estimated attention also remains at zero when we identify it only off the largest downsizings, where a threshold-perception model would predict detection.¹⁶

A related limitation, also concerning how we model attention, is that we capture inattention at the point of purchase and not how it might evolve as consumers gain experience with the downsized products. A consumer who fails to notice the reduced fill in the store may discover it once the package is in use and become attentive to net weight on a later trip, so the inattention we estimate is best interpreted as a short-run, point-of-sale phenomenon. Identifying such learning would require observing repeated purchases of the same product by the same household over a long horizon; given our relatively short sample and the long interpurchase times in the pepper category (a median of 22 weeks between purchases), this is not feasible here. We do, however, find no decline in demand for the downsized products over our sample, which is consistent with limited learning.

Second, the model abstracts from consumer inventory dynamics. With a long shelf life, pepper could be a candidate for stockpiling behavior or dynamic inventory management. The consumer panel data clearly suggest that these concerns are limited. Figure A6 shows the distribution of units purchased per shopping trip, separately for promoted and non-promoted products. Around 89% of trips involving pepper are for non-promoted products, and the vast majority of consumers purchase a single unit regardless of promotional status. Taken together, these patterns are con-

¹⁵Panel B also rejects stability of the price coefficient. This is expected: net weight is mismeasured for the downsized products after 2015, and because price and net weight are correlated, the attenuation in γ spills over to α .

¹⁶As noted in Section VI.B, the value of the content at stake is small enough that attention costs could exceed the benefit even for the largest reductions we observe.

sistent with consumers purchasing pepper as needed rather than stockpiling, which supports the static framework.

V. Estimation Details

A. Market Size

Because the number of potential pepper customers M_{kt} is unobservable, we proxy for it with total unit sales in the seasoning product category at that retailer and month. This product category includes pepper as well as seasoning blends like Old Bay and spices like cinnamon. We do not use the number of visitors to a retailer as many consumers never buy pepper.¹⁷ Given this market size, the share of the outside option is between 69.2 and 97.1%, and 88.5% on average.¹⁸

B. Estimation Procedure

After defining the market size and the shares, we estimate our model of inattention following Berry, Levinsohn and Pakes (1995). In the inner loop, we solve for the mean utilities δ_{jkt} that equate the observed market shares with those predicted by the model; the demand shocks ξ_{jkt} follow once the linear parameters are recovered. In the outer loop, we choose the model parameters to minimize a GMM objective built on the moment conditions $\mathbb{E}[\xi_{jkt}Z_{jkt}] = 0$, given instruments Z_{jkt} . For the instruments, we use a combination of exogenous product characteristics and the local differentiation instruments from Gandhi and Houde (2019). We also include cost shifters, that is, wholesale black- and white-pepper prices and their interactions with package size.

We also allow random coefficients on price and net weight drawn from independent normals with means $\bar{\alpha}$ and $\bar{\gamma}$ and standard deviations σ^p and σ^w , respectively. The inattention parameter follows a Bernoulli distribution where a success indicates that a consumer is attentive. The probability of success is η . As direct estimation of η is numerically unstable near the boundary of $[0, 1]$, we estimate the unconstrained logit transform ζ and recover η via the following logistic transformation, which defines the probability of being attentive as:

$$(7) \quad \eta = \frac{e^\zeta}{1 + e^\zeta},$$

where $\zeta \in \mathbb{R}$ is the unconstrained parameter estimated in the outer loop. The random coefficient τ_i is a draw from this Bernoulli.

The presence of the inattention coefficient τ_i slightly modifies the standard BLP estimation procedure. Normally, one recovers mean utility with the BLP contraction and then recovers the fixed coefficients and the means of the random coefficients using an IV regression. With inattention, $\bar{\gamma}$ enters utility in two places: linearly through $\bar{\gamma}z_{jkt}$ (as usual), and through the wedge term $(1 - \tau_i)\bar{\gamma}\Delta_0 z_{jkt}$ for inattentive consumers, which interacts with the random coefficient τ_i . Because of this nonlinearity, $\bar{\gamma}$ cannot be absorbed into the mean utility δ_{jkt} and must be searched over in the outer loop along with the other nonlinear parameters.

¹⁷About 20% of all panelists present from 2012-2016 have never bought pepper during this period.

¹⁸The high average outside share is consistent with pepper's long interpurchase intervals.

Following BLP, we rewrite the conditional shares in (5) in the following form:

$$s_{ijkt}(\beta, \alpha_i, \tau_i, \gamma_i) = \tau_i \underbrace{\frac{\exp\{\delta_{jkt} + \mu_{ijkt}\}}{1 + \sum_{l \in J_{kt}} \exp\{\delta_{lkt} + \mu_{ilkt}\}}}_{s_{ijkt}^{att}} + (1 - \tau_i) \underbrace{\frac{\exp\{\delta_{jkt} + \mu_{ijkt}^{inatt}\}}{1 + \sum_{l \in J_{kt}} \exp\{\delta_{lkt} + \mu_{ilkt}^{inatt}\}}}_{s_{ijkt}^{inatt}},$$

where

$$\delta_{jkt} = x_{jkt}\beta - \bar{\alpha}p_{jkt} + \xi_{jkt},$$

and

$$\begin{aligned} \mu_{ijkt} &= (\bar{\gamma} + \sigma^w \nu_i^w) z_{jkt} + \sigma^p \nu_i^p p_{jkt} \\ \mu_{ijkt}^{inatt} &= (\bar{\gamma} + \sigma^w \nu_i^w) z_{jkt} + \sigma^p \nu_i^p p_{jkt} + (\bar{\gamma} + \sigma^w \nu_i^w) \Delta_0 z_{jkt}, \end{aligned}$$

with ν_i^w and ν_i^p being draws from standard normals.

Because τ_i takes only two values, the integral over its distribution is the convex combination $\eta s_{ijkt}^{att} + (1 - \eta) s_{ijkt}^{inatt}$, which we compute directly given the closed form.

With this formulation, we apply the modified BLP algorithm. We draw 100 scrambled Halton draws from the distribution of the random coefficients. We simulate the shares and then use the BLP contraction mapping to recover δ . We recover β and $\bar{\alpha}$ from the IV regression of δ_{jkt} on (x_{jkt}, p_{jkt}) . Finally, we find unobserved quality ξ as the residual and construct the sample moments to be minimized with two-step GMM.

To quantify the uncertainty around our estimates, we construct bootstrapped confidence intervals using 250 bootstrap replicates. When constructing the bootstrapped sample, we sample retailer-DMA-month combinations to preserve the structure of the data.

Finally, we perform a Monte Carlo simulation to check the validity of our estimator (Appendix B.B2). Using a data generating process that mirrors our estimated specification, we show that the procedure recovers the true parameter values.

VI. Empirical Results

A. Demand Estimates

Table 2 reports the point estimates and 95% bootstrap confidence intervals across four specifications; all four include the inattention term, differing in which preference parameter is also allowed to be heterogeneous across consumers. Column (1) includes neither price nor net weight as random coefficients; column (2) includes a random coefficient on price; column (3) includes a random coefficient on net weight, but not price; and column (4) adds both.

Table 2—: BLP Demand Estimates with Consumer Inattention

Normal RC on:	(1) None	(2) Price	(3) Net Weight	(4) Both
Means				
Price	-1.607 [-1.657; -1.560]	-1.789 [-1.841; -1.726]	-1.607 [-1.651; -1.565]	-1.774 [-1.828; -1.708]
Net Weight	0.680 [0.657; 0.703]	0.649 [0.629; 0.667]	0.680 [0.661; 0.699]	0.646 [0.626; 0.664]
McCormick	1.951 [1.913; 1.998]	1.989 [1.945; 2.031]	1.951 [1.910; 1.994]	1.987 [1.943; 2.029]
Is Whole	0.126 [0.105; 0.149]	0.066 [0.046; 0.089]	0.126 [0.108; 0.147]	0.063 [0.041; 0.086]
Black Pepper	-0.412 [-0.478; -0.349]	-0.356 [-0.408; -0.295]	-0.412 [-0.466; -0.350]	-0.352 [-0.403; -0.290]
Blend Pepper	-0.168 [-0.207; -0.131]	-0.096 [-0.130; -0.062]	-0.168 [-0.205; -0.133]	-0.093 [-0.126; -0.058]
Cayenne Pepper	-0.496 [-0.557; -0.441]	-0.486 [-0.536; -0.436]	-0.496 [-0.548; -0.446]	-0.482 [-0.534; -0.432]
Citrus Pepper	-2.104 [-2.205; -2.012]	-2.061 [-2.147; -1.966]	-2.105 [-2.191; -2.009]	-2.053 [-2.139; -1.958]
Garlic Pepper	-2.522 [-2.606; -2.445]	-2.498 [-2.576; -2.421]	-2.523 [-2.603; -2.444]	-2.496 [-2.579; -2.419]
Red Pepper	-1.102 [-1.175; -1.037]	-1.126 [-1.190; -1.058]	-1.102 [-1.170; -1.035]	-1.124 [-1.193; -1.057]
Other Pepper	-0.995 [-1.029; -0.959]	-0.987 [-1.020; -0.952]	-0.995 [-1.028; -0.961]	-0.986 [-1.020; -0.952]
Random Coefficients				
Bernoulli: Attention	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]
σ : Price		0.209 [0.187; 0.226]		0.200 [0.176; 0.218]
σ : Net Weight			0.000 [0.000; 0.000]	0.000 [0.000; 0.000]
Observations	1,355,078	1,355,078	1,355,078	1,355,078
Bootstrap replicates	250	250	250	250

Notes: 95% basic-bootstrap confidence intervals in brackets, computed from 250 bootstrap replicates resampled at the retailer-DMA-month level. Reference category for the pepper-type indicators is white pepper. “Is Whole” is an indicator for products containing whole peppercorns. All specifications include a Bernoulli random coefficient on inattention; columns differ in which preference parameters are also treated as normally distributed random coefficients (see Section V.B).

All of the coefficients have the correct expected sign across all specifications. Consumers place a positive valuation on McCormick products. This is not surprising as McCormick is the industry leader and the main brand of pepper in most stores.

Consumers also place a positive valuation on pepper products that contain whole peppercorns. Whole peppercorns stay fresher and more aromatic longer than pre-ground pepper. The grinding process releases some of the essential oils, making pre-ground pepper less fragrant. As a result, products with whole peppercorns are generally of higher quality.

All pepper categories have negative valuations relative to the omitted category, white pepper. Categories like citrus, garlic, and red are niche and less versatile than staple peppers, which explains their large negative coefficients. The negative coefficient on black pepper is more striking: even though black is the most popular variety by volume, white pepper is preferred conditional on price. White pepper requires more processing to remove the outer husk and is therefore more expensive, accounting for its smaller market share despite higher per-unit utility.

As expected, the coefficient on price is negative; consumers prefer to pay lower prices. Specifications

(2) and (4) include a normally distributed random coefficient on price. In both specifications, the standard deviation is positive and significantly different from zero, indicating consumers are heterogeneous in their sensitivity to price.

The coefficient on net weight is positive, indicating that consumers prefer more pepper to less, all else equal. Specifications (3) and (4) include a normally distributed random coefficient on net weight. In contrast to price, the standard deviation on net weight is estimated at zero, indicating no detectable heterogeneity in how consumers value package content. We therefore adopt specification (2), including heterogeneous price preferences only, as our baseline for the counterfactuals that follow. This is the most parsimonious specification that captures all detectable preference heterogeneity in the data. The other coefficient estimates and, critically, the inattention parameter η are essentially identical across all four specifications, so the choice of baseline does not affect our substantive conclusions.

As described previously, the Bernoulli random coefficient represents the probability of being attentive. Across all specifications, the estimated probability of attention is at or near the lower bound of the parameter space, with point estimates of order 10^{-5} or smaller. Consumers are effectively completely inattentive: virtually no one notices downsizing, regardless of which preference parameters are treated as random. Because the estimate lies at the boundary, the bootstrap interval degenerates to $[0.000; 0.000]$ and is uninformative about how large attention could plausibly be. We therefore profile the GMM objective over fixed values of η , re-estimating the remaining parameters at each point. The objective rises steeply away from zero: scaled by the number of observations, its increase exceeds the 5% critical value of a $\chi^2(1)$ at $\eta = 0.00056$. Inverting this test gives an upper bound: the data reject any attention probability above 0.00056, or roughly one consumer in 1,800.

SUBSTITUTION PATTERNS

Translating the price and net-weight coefficients into elasticities shows that consumers respond more strongly to price than to size. Setting the attention parameter to one so that every consumer is fully attentive, the average own-price elasticity is -6.20 and the own-net weight elasticity is 1.96 . Even fully attentive consumers are more than three times as sensitive to a price increase as to an equivalent size decrease. Downsizing therefore remains effective even absent inattention; Kim (2024) also finds that attentive consumers respond less to downsizing than to equivalent price increases. Differential sensitivity to size and price need not, by itself, indicate inattention. At the estimated attention probability, which is essentially zero, the own-net weight elasticity is itself zero while the own-price elasticity is unchanged: almost no consumer reacts to the smaller package at all.

Table 3 reports diversion ratios with respect to net weight, that is, the share of the demand lost by a downsized product that is captured by each competing product. Conditional on switching to another pepper product, about 49% of the diverted demand stays within McCormick & Co, consistent with the estimated brand premium. The composition is nearly unchanged when we set attention to one, and price diversion ratios match the size diversion ratios almost exactly. Inattention therefore suppresses whether consumers respond to a size cut, not where the response goes.¹⁹

¹⁹The estimated diversion is identical to what we obtain if we set each product back to its pre-downsizing package size and make every consumer attentive: since inattentive consumers already perceive the original sizes, they substitute exactly as they would have before the downsizing.

Table 3—: Size Diversion Ratios from Downsized Products

Destination	Estimated η		Full attention ($\eta = 1$)	
	Uncond.	Cond. inside	Uncond.	Cond. inside
Outside good (no purchase)	86.7	—	88.0	—
McCormick, larger (non-downsized) sizes	3.3	25.1	3.5	28.9
McCormick, other downsized sizes	3.2	24.2	2.1	17.7
Other firms	6.7	50.7	6.4	53.4
<i>Own firm (McCormick), total</i>	6.6	49.3	5.6	46.6

Notes: Size diversion ratios from the downsized products, $D_{k \rightarrow j}^{\text{size}} = -(\partial s_j / \partial z_k^p) / (\partial s_k / \partial z_k^p)$, where z_k^p is the net weight the consumer perceives (z_{k0} for an inattentive consumer and z_{kt} for an attentive one). Sales-weighted across downsized products and markets, and evaluated at the estimated η . ‘Cond. inside’ renormalizes over inside goods, i.e. conditional on switching to another pepper product. The two pairs of columns differ because they are evaluated at different choice probabilities: at the estimated η nearly all consumers perceive the original weights, while under full attention all perceive the current ones.

B. Robustness

SELECTION IN DOWNSIZING DECISIONS

In the main specifications, we treat the unobserved demand shock as exogenous to the downsizing characteristic $\Delta_0 z$. Downsizing was triggered by the 2014 spike in wholesale black-pepper prices, an upstream cost shock plausibly exogenous to U.S. retail demand, and was implemented only in February 2015, about a year later. The downsizing episodes are also rare and coarse, clustered at a few focal magnitudes (Section II, Figure A5) rather than finely tuned to each product’s demand.

However, as McCormick downsized a subset of its products, we examine whether that selection was systematic. Table A1 shows that the downsized products had higher pre-downsizing market shares but were statistically indistinguishable on price, package size, and grind type. Because popularity may reflect persistent, product-level unobserved demand, the downsizing magnitude $\Delta_0 z$ may correlate with that component of ξ , biasing the estimates, including η .²⁰

We address this by controlling for the persistent component directly. We proxy it with each product’s pre-shock demand index $\bar{\xi}_j^{\text{pre}}$, the average structural residual from a pre-period Berry (1994) inversion, and add it as a regressor, estimating its coefficient, while price remains instrumented as in the baseline. The index is defined at the product level, where a product j includes both the original and downsized UPC. This control absorbs the persistent demand that may correlate with $\Delta_0 z$ while, unlike product fixed effects, leaving $\Delta_0 z$ and the time-invariant characteristics identified.²¹

Table 4 reports the results. The coefficient on the pre-shock index is positive and somewhat above one, indicating that product-level demand is highly persistent: products with high pre-shock demand remain high-demand over the estimation period, and the control absorbs such persistence. The coefficient of the McCormick indicator decreases in magnitude once persistent demand is controlled for directly, as the pre-shock index absorbs part of the brand’s popularity. Controlling directly for the component that presumably drove the downsizing selection leaves η unchanged.

²⁰In the data, the downsizing magnitude $\Delta_0 z$ is essentially uncorrelated with the pre-shock demand index (slope 0.001, standard error 0.108).

²¹Because $\bar{\xi}_j^{\text{pre}}$ is estimated, we recompute it within each bootstrap replication, so the confidence intervals account for first-stage estimation error.

McCormick’s large share could also reflect heterogeneous brand loyalty: if the consumers purchasing the downsized McCormick products value the brand highly, their failure to switch could bias the estimated attention parameter. We let the taste for McCormick vary with household income, interacting the McCormick dummy with (standardized) income; each market’s consumers draw incomes from their DMA’s distribution in the Nielsen panel, and the interaction π is identified by a micro-moment matching the higher McCormick share among higher-income households.²² Table 4, column (2), reports the estimates. The interaction is positive and precisely estimated ($\pi \approx 0.18$): higher-income consumers value McCormick more. The estimated probability of attention remains very small (0.005), though less precisely estimated than in the main specification, with an upper bound of 0.12. Allowing brand loyalty to vary across consumers therefore leaves the substantive conclusion intact.

Table 4—: Robustness of Inattention to Persistent Demand and Brand Heterogeneity

	(1) Pre-shock demand control	(2) McCormick $\times$ income
Means		
Price	−2.069 [−2.151; −1.992]	−1.636 [−1.693; −1.565]
Net Weight	0.679 [0.650; 0.711]	0.679 [0.647; 0.711]
McCormick	1.429 [1.361; 1.491]	1.954 [1.904; 2.010]
Pre-shock demand index	1.278 [1.259; 1.296]	
No pre-period dummy	−0.455 [−0.516; −0.400]	
Random Coefficients		
Bernoulli: Attention (η)	0.000 [0.000; 0.000]	0.005 [0.003; 0.117]
σ : Price	0.203 [0.180; 0.221]	0.070 [0.047; 0.077]
McCormick $\times$ Income (π)		0.178 [0.171; 0.186]
Bootstrap replicates	250	233

Notes: Point estimates with bootstrapped 95% confidence intervals in brackets. Characteristic coefficients (constant, Is Whole, pepper-type indicators) are estimated but omitted; see Table 2. Column (1) controls for persistent product-level demand by adding each product’s pre-shock demand index $\hat{\xi}_j^{\text{pre}}$ (the average Berry (1994) residual from a pre-period inversion, pooling a product’s original and downsized UPCs) as a regressor; the “no pre-period” dummy flags families lacking pre-downsizing data, and $\hat{\xi}_j^{\text{pre}}$ is recomputed within each replicate so the intervals reflect first-stage estimation error. Column (2) interacts the McCormick dummy with standardized household income, drawn from each market’s DMA-level income distribution in the Nielsen consumer panel, with π identified by a micro-moment on the McCormick share of higher- versus lower-income households. Column (2) excludes 17 of 250 replicates where the optimizer failed to converge.

²²The moment is the gap in the average within-household McCormick share of pepper purchases between above- and below-median-income households (0.072, $t \approx 13$). Income is standardized, so π is the effect of a one-standard-deviation rise in income on McCormick utility. Results are similar with an unobserved normal random coefficient on the McCormick dummy, which we omit because it implies very large taste dispersion and absorbs the price random coefficient.

ALTERNATIVE UTILITY SPECIFICATIONS

One concern is that the estimated inattention probability may depend on how net weight enters utility. In the main specifications, we assume net weight enters linearly. The functional form matters because it determines the utility wedge from inattention; Appendix B.B3 derives the wedge under: (i) a quadratic specification; (ii) a log specification; and (iii) a specification in which net weight is interacted with the indicator denoting whole pepper; this interaction is motivated by the observation that whole peppercorns differ from pre-ground varieties (specifically in terms of longer shelf life), so consumers who select whole peppercorns may have systematically different preferences for weight. As a robustness check, we re-estimate the model under each of these three alternatives. Columns (1) to (3) of Table 5 report the estimates under each alternative in turn.

The functional form for net weight does not affect the main results. All of the non-size coefficients have the same sign and similar magnitude to those in our main specification. Under all specifications, consumers prefer more pepper to less, all else equal.²³ Column (1) replaces the linear net-weight term with a quadratic one. The linear coefficient on net weight is large and positive ($\gamma_1 = 0.87$) while the quadratic coefficient is small and negative ($\gamma_2 = -0.015$), indicating concavity: consumers value additional pepper, but the marginal benefit declines slightly with package size. The inattention wedge becomes $\gamma_1 \Delta_0 z_{jkt} + \gamma_2 \Delta_0(z_{jkt}^2)$, where $\Delta_0(z_{jkt}^2) = z_{jk0}^2 - z_{jkt}^2$, which adjusts the welfare cost of inattention by a second-order term in net weight.

Column (2) replaces net weight with its logarithm in utility. The coefficient on $\log(z)$ is $\gamma = 2.595$, substantially larger in magnitude than the linear-specification estimate because the log transformation compresses variation in package size. The inattention wedge, $\gamma \cdot (\log z_{jk0} - \log z_{jkt})$, depends only on the percentage change in net weight rather than its absolute level.

Column (3) introduces the interaction between net weight and an indicator for whole peppercorns. The base net-weight coefficient is $\gamma = 0.597$ and the interaction coefficient is positive and precisely estimated ($\phi = 0.369$), indicating that consumers buying whole peppercorns value an additional ounce of pepper roughly $\phi/\gamma \approx 62\%$ more than consumers who buy pre-ground varieties. Allowing this heterogeneity changes the inattention wedge to $(\gamma + \phi w_j) \Delta_0 z_{jkt}$, where w_j indicates whole peppercorns, so consumers of whole-peppercorn products experience a larger utility shift from a given net-weight reduction.

Most notably, across all three alternatives, the estimated probability of being attentive remains zero.

IDENTIFYING VARIATION DURING THE OVERLAP PERIOD

As discussed in Section IV.C, a possible concern is that the overlap between the original and downsized products may bias the probability of being attentive toward zero. Because we aggregate the scanner data to the retailer, DMA, and month level, the original and downsized product can appear in the same choice set despite never being sold in the same store at the same time. For a given retailer and DMA, one store may stock the original product, while the others stock the downsized product. During the transition from the original product to the downsized product, the share of the original product will decrease as the share of the downsized product increases, potentially creating

²³In the quadratic specification, the marginal utility of net weight is $\gamma_1 + 2\gamma_2 z = 0.869 - 0.030z$, which is positive up to 29 ounces. Only 0.02 percent of observations involve packages larger than that, so the marginal utility is positive over essentially the entire support of package sizes.

bias. In particular, observing market shares that gradually shift from the original to the downsized product could be misread as evidence of inattention, even when consumers in any individual store only ever faced one of the two versions. This would bias the estimated probability of being attentive downward.

To test whether the overlap biases our results, we estimate our model on (i) a sample that excludes the overlap and (ii) a sample consisting only of the overlap. Columns (4) and (5) of Table 5 show the BLP results for the non-overlap and overlap samples. The probability of being attentive is zero in both samples, so the overlap does not drive our result.

Table 5—: Robustness: Alternative utility specifications and Overlap period

	(1)	Functional Form (2)	(3)	No Overlap (4)	Overlap (5)
Means					
Price	-1.741 [-3.251; -1.690]	-2.210 [-2.260; -2.157]	-1.866 [-1.921; -1.798]	-1.811 [-1.876; -1.762]	-1.463 [-1.640; -1.379]
Net Weight	0.869 [0.822; 1.738]		0.597 [0.579; 0.615]	0.716 [0.691; 0.739]	0.421 [0.382; 0.458]
(Net Weight) ²	-0.015 [-0.029; -0.012]				
log(Net Weight)		2.595 [2.552; 2.637]			
Whole $\times$ Net Weight			0.369 [0.356; 0.382]		
McCormick	1.894 [1.853; 2.813]	1.754 [1.722; 1.782]	2.006 [1.961; 2.049]	2.036 [1.977; 2.093]	1.773 [1.722; 1.850]
Random Coefficients					
Bernoulli: Attention	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]
σ : Price	0.233 [0.216; 0.466]	0.511 [0.492; 0.528]	0.193 [0.170; 0.213]	0.174 [0.155; 0.195]	0.254 [0.208; 0.470]
Observations	1,355,078	1,355,078	1,355,078	835,751	519,327
Bootstrap replicates	232	250	250	250	250

Notes: Point estimates with bootstrapped 95% confidence intervals in brackets. Characteristic coefficients (constant, Is Whole, pepper-type indicators) are estimated but omitted; see Table 2. Specification (1) excludes replicates where the optimizer failed.

A THRESHOLD-PERCEPTION FRAMEWORK

A threshold-perception framework would predict that consumers notice sufficiently large reductions while overlooking small ones, so that measured inattention should fall as the magnitude of the downsizing rises. We test this by re-estimating the model with the inattention wedge active only for downsizings above a threshold, treating consumers as attentive to smaller reductions. We apply two thresholds: a *relative* threshold retaining downsizing of at least 25% of the original size, and an *absolute* threshold retaining downsizing of at least one ounce.²⁴ As Table 6 shows, the estimated

²⁴Each case zeroes the perceived-size wedge $\bar{\gamma} \Delta_0 z$ for products below the threshold, so η is identified only off the products downsized above it. The two thresholds retain different products: the absolute one keeps the large jars whose reductions are sizeable in ounces, but in some cases modest in percentage terms; the relative threshold excludes these.

probability of attention remains zero under both scenarios. Consumers fail to respond even to the largest downsizings, whether “large” is measured proportionally or in absolute content. These results do not fully rule out rational inattention: at the estimated preferences, the content lost to a downsizing is worth between \$0.09 and \$0.72 per purchase over the observed range of reductions, and pepper is bought less than once a year on average, so attention costs may exceed the benefit for every downsizing we observe, whatever its magnitude. For the counterfactuals the distinction is immaterial: a disclosure mandate reduces the cost of attention to effectively zero.

UNIT-PRICE LABELING REQUIREMENTS

If inattention stemmed from the difficulty of comparing unit prices, it should be weaker where unit-price disclosure is mandated. We estimate the model separately for markets in states that require unit-price disclosure (CT, MD, MA, NH, NJ, NY, OR, RI, VT, and DC) and in those that do not. The estimated attention probability is zero in both groups (Table 6); we conclude that inattention to downsizing mainly operates through channels other than the salience of per-unit prices.²⁵ While Wan, Harris-Lagoudakis and Crespi (2026) find that consumers in disclosure states respond more strongly to the unit-price increases that downsizing produces in canned tuna, we find no such difference for pepper. Their evidence that information provision changes behavior is nonetheless consistent with our reading of the non-response as informational rather than preference-based.

Table 6—: Robustness: Large Downsizings and Unit-Price Disclosure

	Large downsizings		Unit-price disclosure	
	$\geq 25\%$	≥ 1 oz	Mandate	No mandate
Means				
Price	-2.090 [-2.147; -2.029]	-1.992 [-2.051; -1.930]	-1.313 [-1.376; -1.260]	-1.932 [-1.989; -1.880]
Net Weight	0.692 [0.674; 0.712]	0.676 [0.657; 0.694]	0.631 [0.609; 0.661]	0.604 [0.586; 0.630]
McCormick	1.853 [1.809; 1.893]	2.003 [1.956; 2.048]	1.141 [1.064; 1.223]	1.983 [1.948; 2.027]
Random Coefficients				
Bernoulli: Attention	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]	0.000 [0.000; 0.000]
σ : Price	0.286 [0.268; 0.303]	0.255 [0.237; 0.270]	0.000 [0.000; 0.000]	0.321 [0.301; 0.337]
Observations	1,355,078	1,355,078	86,825	1,144,859
Bootstrap replicates	249	250	250	250

Notes: 95% basic-bootstrap confidence intervals in brackets, resampled at the retailer-DMA-month level. Characteristic coefficients (constant, Is Whole, pepper-type indicators) are estimated but omitted; see Table 2. In the “Large downsizings” columns the perceived-size wedge is set to zero for downsizings below the threshold, so attention is identified only off reductions of at least 25% (relative) or at least one ounce (absolute). The “Unit-price disclosure” columns restrict to markets in states that mandate unit-price disclosure (CT, MD, MA, NH, NJ, NY, OR, RI, VT, DC) or that do not. Nielsen designated market areas can span more than one state. The two columns retain only DMAs lying fully within mandate or fully within non-mandate states; 123,394 observations in DMAs that straddle the two are excluded from both.

²⁵Unit-price labels support comparisons across sizes at a point in time; they do not signal that a product’s content has changed, which is what the French and Austrian rules require, and what our first counterfactual evaluates.

VII. Counterfactuals

In this section, we consider the welfare implications of consumer inattention to downsizing. Inattention can reduce welfare by distorting product choices, as consumers maximize perceived utility rather than actual utility. While, in the long run, consumers can presumably learn about downsizing and adjust their choices accordingly, nudges and notices can help consumers make more informed choices and combat inattention in the short run.

Reducing inattention does not address the direct welfare loss from smaller products that yield less utility. Restricting firms from downsizing prevents both inattention-related issues and utility loss from smaller packages. However, without downsizing, firms will raise unit prices by increasing package prices, which may offset the gain from preserving larger sizes.

To examine these trade-offs, we conduct two counterfactual simulations. In the first counterfactual, we quantify the welfare loss due to inattention. We assume that firms still engage in downsizing, but that consumers are now attentive. The second counterfactual examines whether consumers would be better off if downsizing had never occurred. These counterfactual scenarios provide guidance to policymakers about the benefits of improving attention to downsizing or banning downsizing entirely.

Removing inattention and banning downsizing both change equilibrium prices, which we recompute from the pricing first-order conditions in equation (10). In the first scenario, package sizes are unchanged, so the marginal costs recovered from equation (10) remain unchanged and we do not need to estimate a cost function in characteristics. The second scenario changes net weights; we therefore model and estimate how marginal cost responds to net weight, adjust costs when products are restored to their original weights, and solve for the new price equilibrium. We describe the supply specification before discussing the results of the counterfactuals.

A. Supply

We assume that manufacturers simultaneously set retail prices and retailers extract profits through slotting fees. To maximize their profits, manufacturers set the wholesale price equal to the retail price, so the retail margin is zero.

Because consumers choose retailers based on a basket of goods, pepper prices are unlikely to affect consumers' choice of retailer. Consequently, each retailer–DMA can be treated as a separate market: manufacturers maximize profits retailer by retailer, and there is no substitution across retailers in response to pepper prices. A manufacturer m sets the retail price at a retailer–DMA k in period t to maximize its profits:

$$(8) \quad \pi_{kt}^m = \sum_{j \in J_{mkt}} \left[p_{jkt} - mc_{jkt} \right] s_{jkt}(p_{kt}) M_{kt}$$

where J_{mkt} is the set of products that m sells to retailer k at time t ; M_{kt} is the market size; and mc_{jkt} is the manufacturer's marginal cost.

The first-order condition with respect to p_{jkt} is:

$$(9) \quad s_{jkt} + \sum_{r \in J_{mkt}} \left[p_{rkt} - mc_{rkt} \right] \frac{\partial s_{rkt}}{\partial p_{jkt}} = 0.$$

Stacking the first-order conditions of all manufacturers and rearranging terms, the equilibrium prices satisfy:

$$(10) \quad p_{kt} + \Delta_{kt}^{-1} s_{kt}(p_{kt}) = mc_{kt}$$

where Δ_{kt} is the ownership matrix with entry (r, s) equal to $\frac{\partial s_{rkt}}{\partial p_{skt}}$ if a single manufacturer sells both products r and s to retailer k during period t , and zero otherwise.²⁶

From the first-order conditions, we recover the marginal costs. We assume a specification where the marginal cost of a product j is a quadratic function of its net weight z_{jkt} , together with other observable characteristics x_{jkt} and an unobserved cost shock ω_{jkt} :

$$(11) \quad mc_{jkt} = [p_{kt} + \Delta_{kt}^{-1} s_{kt}(p_{kt})]_j = x_{jkt}\kappa_1 + z_{jkt}\kappa_2 + z_{jkt}^2\kappa_3 + \omega_{jkt}$$

The quadratic term allows the cost of an additional ounce to depend on how large the package is. We expect $\kappa_2 > 0$ and $\kappa_3 < 0$, reflecting economies of scale in filling, packing, and distribution. We find that marginal cost rises with net weight over essentially the whole range of package sizes in our sample, but at a declining rate. The linear restriction $\kappa_3 = 0$ is rejected. Appendix B.B4 reports the full set of functional forms we considered and shows that every specification permitting curvature implies a larger cost of restoring package size than the quadratic, so the counterfactual in Section VII is conservative.

We estimate the marginal cost parameters by ordinary least squares. The estimated parameters allow us to adjust marginal costs in the case of a ban on downsizing, when firms cannot adjust net weights. Crucially, marginal costs increase with net weight, so firms that downsize face lower costs. We take firms' product assortment, including package sizes, as given and model only the pricing stage. Package size is a discrete characteristic that is costly to adjust: changing it requires retooling packaging at a sunk cost, and firms choose from a limited menu of standard formats rather than a continuum.

The downsizing episodes we observe are consistent with this description: reductions are rare and cluster at a few focal magnitudes, with gaps in between (Section II, Figure A5). Because size adjustments are coarse and infrequent, modeling the size decision as a smooth response to demand would impose more structure than the data support. Taking the assortment as given is also sufficient for our purposes: both counterfactuals condition on the observed set of products. The first re-prices the existing products under full attention, and the second restores their original weights. We therefore never need to predict which products a firm would choose to downsize.

Holding the assortment fixed also shapes how the first counterfactual should be read. We raise attention to one while keeping the product set unchanged, so every downsized product remains in the choice set and is fully noticed. A firm facing attentive consumers, however, could choose a different assortment, and by construction its profits under that assortment are at least as high as under the observed one. The loss in demand and profits we compute therefore overstates the value

²⁶Private-label products are attributed to the retailer that sells them, so each chain's store-brand line is a single pricing entity.

the firm extracts from inattention: allowing re-optimization can only raise attentive-world profits and narrow the gap. This holds whichever products the firm would choose to downsize, so we do not need to take a stand on that. We accordingly read the counterfactual as a bound rather than a point prediction, which also makes the policy conclusion robust to the concern that firms selectively downsized their most popular products (Section VI.B). The bound is short-run, consistent with our interpretation of inattention as a point-of-sale phenomenon: over a longer horizon, consumers could learn about the reductions. A dynamic model of assortment choice in the spirit of Wollmann (2018) would require specifying both the horizon over which consumers learn and the rate at which they do so; the data offer little guidance on either. The static model avoids these assumptions and delivers the bound directly.

B. First Counterfactual: Removing Inattention

EQUILIBRIUM PRICES AND QUANTITIES

In this counterfactual we ask how market outcomes and consumer welfare would change if consumers were fully attentive, holding the set of products fixed, as discussed in Section VII.A. We set $\tau_i = 1$ for every consumer (equivalently, $\eta = 1$), so that perceived utility coincides with actual utility and the inattention wedge $\gamma_i \Delta_0 z_{jkt}$ is removed. Because inattention distorts perceived utilities, removing it shifts demand and therefore equilibrium prices and quantities.

Removing inattention acts as a quality reduction for the downsized products: consumers, who previously evaluated them at their original net weight, evaluate them now at their current one and find them less attractive. Demand shifts away from these products and toward the rest of the choice set. This puts downward pressure on the prices of the firms that downsized and upward pressure on those of rival firms, although the movement in any given market depends on ownership structure and substitution patterns. We denote the resulting equilibrium prices by p_{kt}^a .

Under full attention, demand $s_{kt}^a(\cdot)$ no longer depends on the size change, and the matrix of price responses Δ_{kt}^a , with entry (r, s) equal to $\partial s_{rkt}^a / \partial p_{skt}$, is evaluated at the new demand system. Crucially, attention does not enter the cost side: manufacturers' marginal costs are unchanged. Counterfactual prices therefore satisfy

$$(12) \quad p_{kt}^a + (\Delta_{kt}^a)^{-1} s_{kt}^a(p_{kt}^a) = p_{kt} + \Delta_{kt}^{-1} s_{kt}(p_{kt}) = mc_{kt},$$

which defines the counterfactual prices implicitly as a function of the demand-side parameters alone.

We first recover marginal costs from observed prices using the second equality in equation (12). To solve for the counterfactual prices, we follow Morrow and Skerlos (2011) and Conlon and Gortmaker (2020) and rewrite the markup equation as a contraction mapping, on which we then iterate.

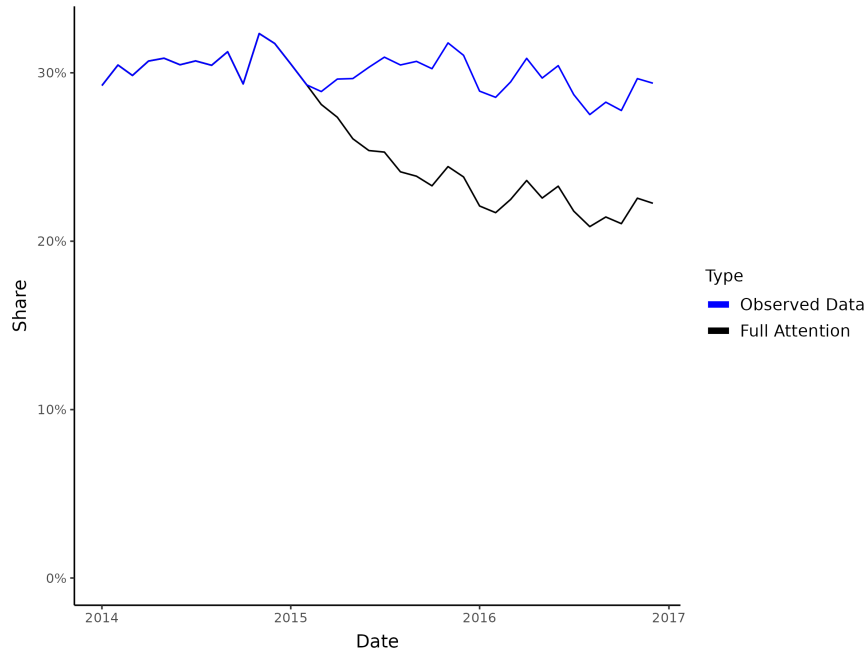
Figure A7 shows the difference between the observed and the counterfactual prices for all products from 2015 onward, decomposed by ownership, and Figure A8 shows the distribution for the downsized products alone. In 11% of observations the price does not move because the retailer does not stock a downsized product.

The price adjustments are uniformly small. No product changes price by more than \$0.05 and 90% change by less than a cent; the direction of the change is driven almost entirely by ownership. Prices fall for 98.7% of the downsized products and for 98.2% of the non-downsized products sold

by the same firms, while they rise for every product sold by an independent competitor. Relative to the package price, the downsized products fall by 0.20% and the non-downsized products of the same firms by 0.21%.

In a multiproduct setting, equilibrium markups depend largely on a firm's aggregate share rather than on the share of individual products. Removing inattention reduces the share of the firms that downsized, so their markups fall across their entire portfolios, while their competitors gain share and raise prices. However, because the implied changes in share are small, the markup adjustments are also small in absolute terms. Inattention thus affects welfare mainly through the products consumers choose rather than through the prices they pay.

Figure 5. : Counterfactual 1: Market Shares



Notes: Share of the downsized products among the inside goods, weighted by units sold. The blue line is the observed share; the black line is the share when all consumers are attentive. The two coincide through 2014, when no product has yet been downsized.

Figure 5 reports the share of the downsized products among the inside goods, averaging across retailer–DMA markets in proportion to the units they sell. The two series coincide through 2014, when no product has been downsized, and diverge from 2015 onward as more products are reduced in size. Across all 34,056 retailer–DMA–months the share of the downsized products falls from 31.5 to 27.6% under full attention. Restricting attention to the 18,628 markets that actually contain a downsized product, it falls from 31.2 to 24.1%, a decline of 7.1 percentage points or 22.6%. The effect grows as downsizing spreads: 14.3% in 2015 and 25.3% in 2016. Attentive consumers substitute away from downsized packages at a substantial rate.

The response is concentrated in the downsized products rather than in the firms selling them. In markets containing a downsized product, the share of the inside goods held by McCormick falls from 41.3 to 38.3%, a decline of 3.1 percentage points or 7.5%, against 22.6% for the downsized products themselves. Most of the demand that leaves a downsized package is thus recaptured

by other products in the same firm's line rather than by rivals or by the outside option. Still, McCormick's variable profits fall by 10.1%, or \$3.8 million over the three years of our sample. Inattention is therefore profitable for the firm.

CONSUMER WELFARE

Inattentive consumers buy a downsized product under the belief that it contains more pepper than it actually does. After purchasing, some of these consumers may experience regret when they discover the smaller package content. Post-purchase regret, however, does not necessarily imply welfare losses. For inattention to reduce consumer welfare, it must alter the choices consumers make or allow firms to charge higher prices.

Because inattention distorts perceived utilities, some inattentive consumers choose a product that is not utility maximizing. Only inattentive consumers can experience this type of loss. Inattention also affects consumers indirectly through prices. Firms that downsize hold larger shares when consumers do not notice, and charge higher markups across their product lines as a result; their competitors charge less. The net effect on any individual consumer therefore depends on what she buys. The price channel reduces consumer surplus in 81% of the markets containing a downsized product and raises it in the remaining 19%, where the basket is weighted toward firms that did not downsize and that price lower when consumers are inattentive.

Consumers can choose the wrong product because they base their purchase decision on perceived rather than actual utility. An inattentive consumer at retailer k in period t chooses

$$j^* = \arg \max_{0, \dots, J_{kt}} U_{ijkt}^p(p_{kt}),$$

while the product that would in fact serve her best at those same prices is

$$m^* = \arg \max_{0, \dots, J_{kt}} U_{ijkt}^a(p_{kt}).$$

Comparing j^* with m^* would hold prices fixed. Removing inattention shifts demand and therefore moves equilibrium prices, so an attentive consumer would face p_{kt}^a rather than p_{kt} . The relevant benchmark is the choice she would make if she were attentive and faced the counterfactual prices:

$$c^* = \arg \max_{0, \dots, J_{kt}} U_{ijkt}^a(p_{kt}^a).$$

The utility she loses to inattention is therefore:

$$(13) \quad \mathcal{W} = U_{ic^*kt}^a(p_{kt}^a) - U_{ij^*kt}^a(p_{kt})$$

Note that j^* and c^* depend on the random parameters and the taste shock ε_{ijkt} . Taking the expectation over these gives the average welfare loss from imperfect attention:

$$(14) \quad \Delta CS = \mathbb{E} \left[\frac{\mathcal{W}}{\alpha_i} \right] = \mathbb{E} \left[\frac{U_{ic^*kt}^a(p_{kt}^a)}{\alpha_i} \right] - \mathbb{E} \left[\frac{U_{ij^*kt}^a(p_{kt})}{\alpha_i} \right]$$

Given the logit form, the welfare loss is:

$$\begin{aligned}
 \Delta CS = \int & \left[\log \left(1 + \sum_{j \in J_{kt}} \exp \{ x_{jkt} \beta - \alpha_i p_{jkt}^a + \xi_{jkt} + \gamma_i z_{jkt} \} \right) \right. \\
 (15) \quad & - \log \left(1 + \sum_{j \in J_{kt}} \exp \{ x_{jkt} \beta - \alpha_i p_{jkt} + \xi_{jkt} + \gamma_i z_{jkt} + (1 - \tau_i) \gamma_i \Delta_0 z_{jkt} \} \right) \\
 & \left. + \sum_{j \in J_{kt}} s_{ijkt}(\beta, \alpha_i, \tau_i, \gamma_i) (1 - \tau_i) \gamma_i \Delta_0 z_{jkt} \right] \frac{1}{\alpha_i} dG(\tau, \alpha, \gamma)
 \end{aligned}$$

where s_{ijkt} is the choice probability in equation (5), evaluated at perceived utility and observed prices. Our formula for the welfare loss is close to that of Train (2015); we modify the equation to incorporate price changes due to the removal of imperfect attention, as in Stivers (2019).²⁷

The equation has three components. The first term is the standard log-sum formula based on actual utility evaluated at the counterfactual prices, where the log-sum is the closed-form expression for expected maximum utility. The second term is the log-sum formula based on perceived utility at the observed prices; it is the expected utility the consumer believes she obtains, which exceeds what she actually obtains. The final term corrects for that gap: it is the average amount by which perceived utility exceeds actual utility at the chosen product. This term is a summation over the downsized products alone, since size does not change for the non-downsized products.

Consumers can experience a loss in welfare only at retailers that offer downsized products; if a retailer does not stock a downsized product, consumers cannot mistakenly choose one. Of the 34,056 retailer–DMA–months in our sample, 18,628 contain at least one downsized product. Restricting to the 22,050 market–months observed in 2015 and 2016, after downsizing begins, the share is 84%. Most market–months after downsizing therefore expose consumers to a potential welfare loss.

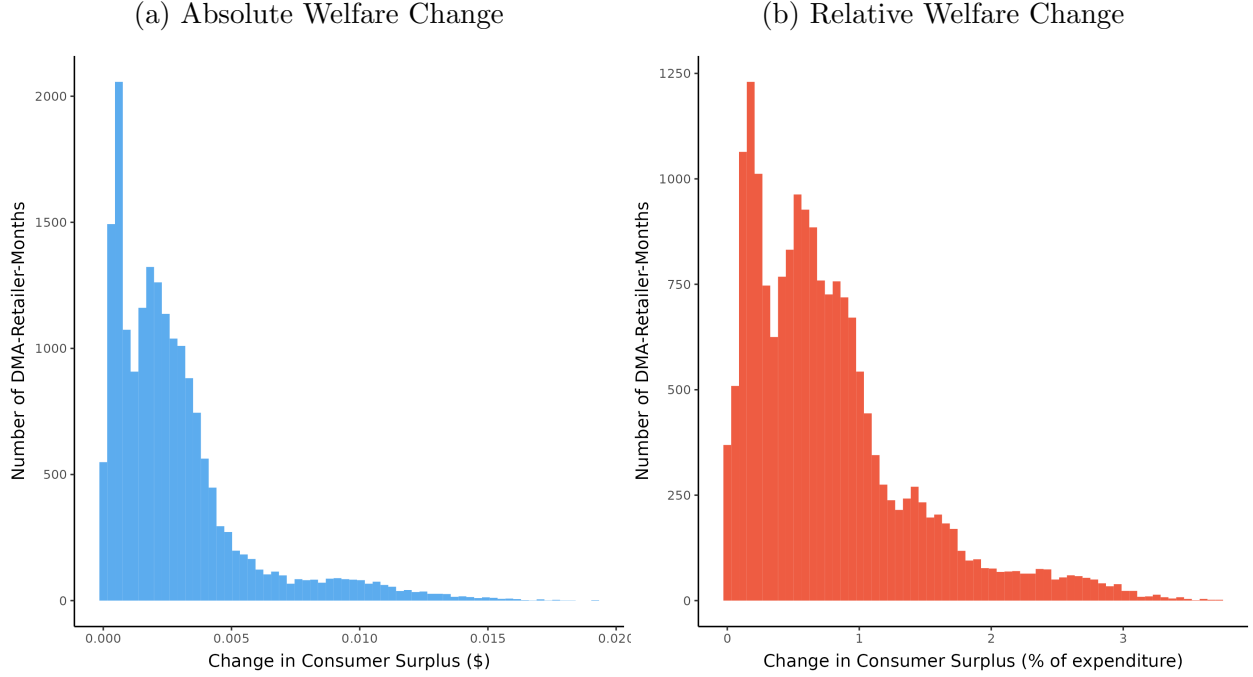
Figure 6a shows the histogram of absolute welfare losses in markets with downsized products. In absolute terms the loss is modest for most consumers: it ranges from essentially zero to \$0.020 per consumer, with a mean of \$0.0028 and a median of \$0.0022.

Because pepper is inexpensive, the relative loss is easier to interpret. Figure 6b expresses the loss as a percentage of per-capita expenditure on pepper in the same retailer–DMA–month. It averages 0.78% in markets containing a downsized product, with a median of 0.64%, and reaches 3.8% in the upper tail. Aggregated over the sample, inattention costs consumers \$1.84 million across the three years, or roughly \$613,000 a year. This is 8.2 cents per downsized package sold, or about 2.3% of the \$3.61 average price of a downsized package.

Prices account for a modest part of this. Holding prices at their observed values, the loss would be 0.685% rather than 0.779%, so roughly 12% of the total operates through prices and the remaining 88% through consumers choosing the wrong product because of inattention.

²⁷We report consumer surplus conditional on $\alpha_i > 0$, since a normal price coefficient places positive density at zero and the unconditional expectation of $1/\alpha_i$ is not defined. At our estimates, $\Pr(\alpha_i \leq 0)$ is approximately zero, so the conditioning does not affect the reported figures.

Figure 6. : Welfare Change from Removing Inattention



Notes: Each observation is a retailer–DMA–month containing at least one downsized product (18,628 markets); markets with no downsized product have a welfare change of exactly zero and are excluded. Panel (a) reports the change in consumer surplus per consumer, in dollars, and panel (b) the same change as a percentage of per-capita expenditure on pepper in that market and month. Positive values indicate a gain; removing inattention leaves no market worse off. Both histograms are trimmed at the 0.01st and 99.99th percentiles, so the extreme upper tail is not visible.

C. Second Counterfactual: Banning Downsizing

EQUILIBRIUM PRICES AND QUANTITIES

We now consider how consumer welfare would change if manufacturers were prohibited from reducing package sizes. We can think of this counterfactual as a ban on downsizing. We assume that manufacturers keep their product offerings the same, but offer the original versions in place of the downsized ones.

Swapping the downsized version for the original one affects both demand and supply. The greater net weights on some products change consumers’ utilities and therefore demand. Because there is no downsizing, however, inattention no longer affects demand, as there is nothing for consumers to miss, and the wedge $\Delta_0 z_{jkt}$ is zero.

The increase in net weight raises the marginal costs of the affected products, so we use the supply-side estimates to adjust marginal costs to the original net weights. Substituting z_{jk0} into equation (11), the marginal cost of a restored product rises by $\kappa_2(z_{jk0} - z_{jkt}) + \kappa_3(z_{jk0}^2 - z_{jkt}^2)$, resulting in the new marginal costs $mc_{kt}(z_{k0})$. Table B3 reports that, across the downsized products, this adjustment averages \$0.36 per package. The new equilibrium prices p'_{kt} satisfy:

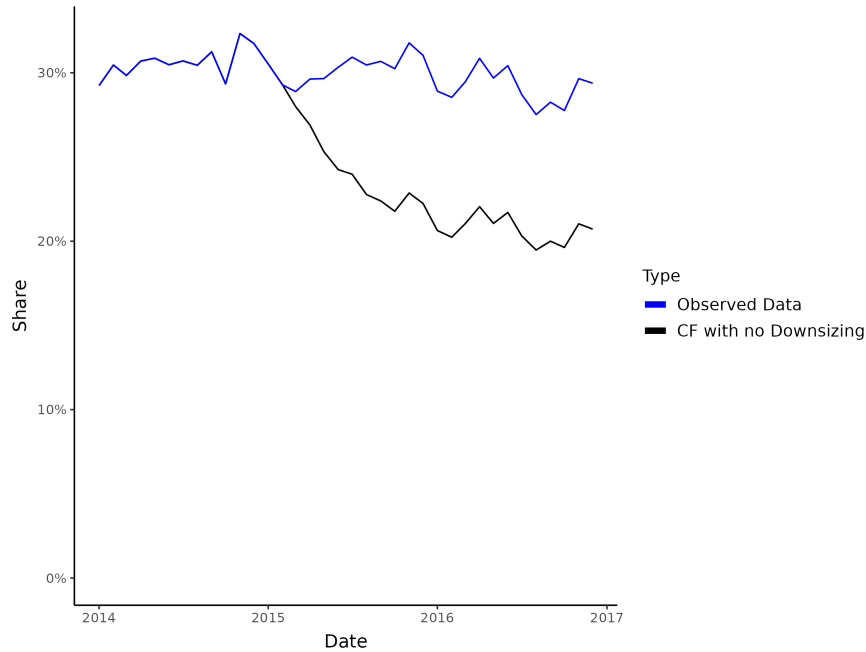
$$(16) \quad p'_{kt} + (\Delta'_{kt})^{-1} s'_{kt}(p'_{kt}) = mc_{kt}(z_{k0})$$

where Δ'_{kt} and s'_{kt} are evaluated at the restored net weights and the new prices.

Figure A9 shows that products sold by the same firms, and by their competitors, change price by less than a cent on average. While, in the first counterfactual, the removal of inattention lowered prices across the entire portfolios of the firms that had downsized, a ban instead raises the price of specific products and leaves the rest of the market almost untouched. Figure A10 shows the price changes for the downsized products. Every product becomes more expensive, by \$0.36 on average, or 9.6% of its price, with half of the increases falling between \$0.26 and \$0.49 and 3.4% exceeding \$0.80.

The downsized products lose a substantial share of the market under the ban. In the 18,628 markets containing a downsized product, their share of the inside goods falls from 31.2 to 22.8%, a decline of 8.4 percentage points or 27.0%, against 22.6% under full attention. The effect grows over time: 17.3% in 2015 and 30.0% in 2016 (Figure 7).

Figure 7. : Counterfactual 2: Market Shares



Notes: Share of the downsized products among the inside goods, weighted by units sold. The blue line is the observed share; the black line is the share under a ban on downsizing, with the original net weights restored and marginal costs adjusted using the quadratic specification of equation (11).

Consumer welfare is computed as before. The welfare loss takes the form:

$$\begin{aligned}
 \Delta CS = & \int \left[\log \left(1 + \sum_{j \in J_{kt}} \exp \{ x_{jkt} \beta - \alpha_i p'_{jkt} + \xi_{jkt} + \gamma_i z_{jk0} \} \right) \right. \\
 (17) \quad & - \log \left(1 + \sum_{j \in J_{kt}} \exp \{ x_{jkt} \beta - \alpha_i p_{jkt} + \xi_{jkt} + \gamma_i z_{jkt} + (1 - \tau_i) \gamma_i \Delta_0 z_{jkt} \} \right) \\
 & \left. + \sum_{j \in J_{kt}} s_{ijkt}(\beta, \alpha_i, \tau_i, \gamma_i) (1 - \tau_i) \gamma_i \Delta_0 z_{jkt} \right] \frac{1}{\alpha_i} dG(\tau, \alpha, \gamma),
 \end{aligned}$$

where the first log-sum now uses the original net weights z_{jk0} and the equilibrium prices under the ban, p'_{kt} . The second and third terms are unchanged, as they describe the world we observe. As mentioned above, we adopt the quadratic cost specification, which fits better than the linear one and whose concavity reflects economies of scale in filling, packing, and distribution.²⁸

Figure 8a shows the histogram of absolute welfare changes in markets containing a downsized product. For most consumers, banning downsizing produces a small gain: the median is \$0.0010 per consumer and the mean \$0.0017, with an upper tail reaching \$0.027. Unlike the removal of inattention, however, the distribution extends below zero. In relative terms, the ban raises consumer surplus by 0.43% of per-capita category expenditure in markets with a downsized product, against 0.78% for the removal of inattention through disclosure, and the median gain is 0.30% against 0.64%. Aggregated over the sample, the ban is worth \$982,000 across three years, or \$327,000 a year, compared with \$1.84 million and \$613,000 for disclosure. A ban therefore delivers a little over half the gain of making consumers attentive. It also makes some consumers worse off, while disclosure never does: in 2.5% of markets containing a downsized product, the price increase exceeds the value of the restored content.

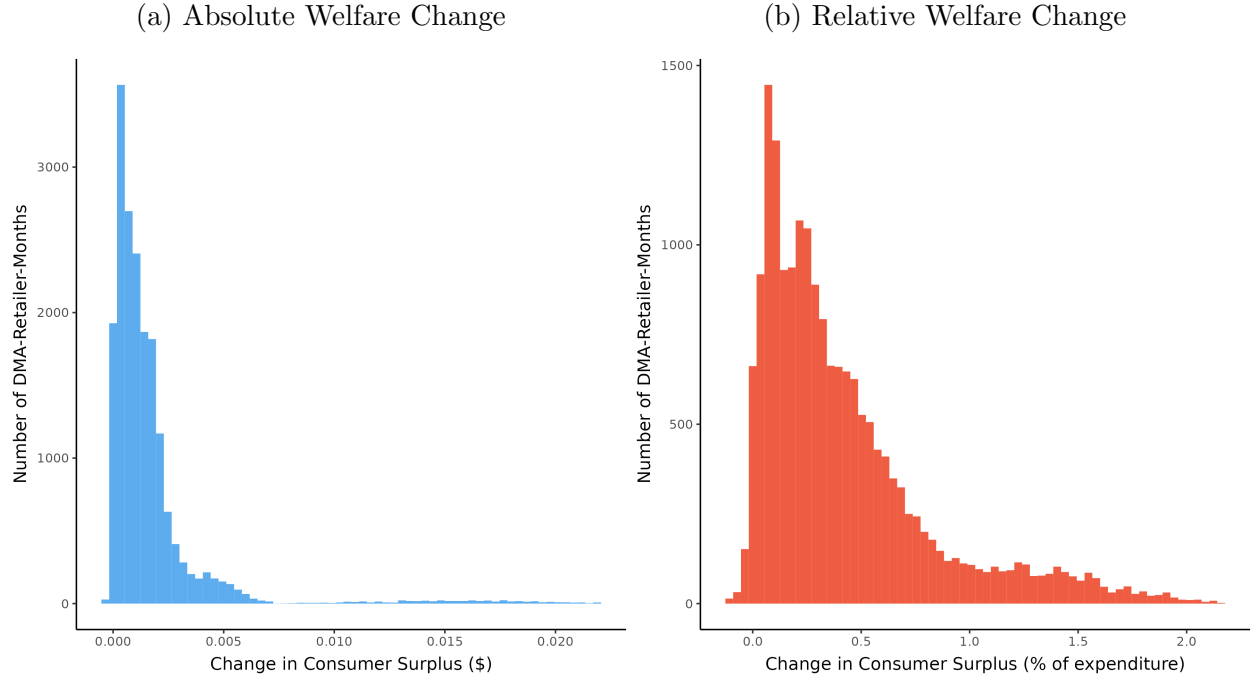
A ban underperforms because it tackles inattention only indirectly. Restoring the original net weights solves choice distortions arising from inattention because consumers have nothing to overlook, but also raises marginal costs by \$0.36 per restored package, which firms pass through almost fully into prices. The average restored package gains 0.76 ounces, so the ban supplies content at roughly \$0.48 per ounce while consumers value an ounce at $\gamma/\alpha = \$0.36$: because they are more sensitive to package price than to net weight, the restored content is worth less to them than it costs to provide. Removing inattention corrects the same choice distortion without imposing that cost.

The magnitude of the gain from a ban depends on the cost specification; its ranking against removing inattention does not. Under the logarithmic specification favored by the encompassing test in Appendix B.B4, the cost of restoring a package is \$0.44 rather than \$0.36, and that difference is enough to reverse the sign: a ban then *lowers* consumer surplus by 0.04% on average and leaves 65% of affected markets worse off. Removing inattention does not depend on the cost specification. Across all cost specifications the data admit, removing inattention raises consumer surplus by more than a ban does. A disclosure rule could in principle act like a ban; firms could respond to disclosure

²⁸Because the estimated curvature is concave, marginal cost eventually turns down, at 27.0 ounces. That point lies inside the 0.4–32 ounce estimation range but above nearly all of the data: only 328 observations, 0.02 percent of the sample, exceed it, and the largest downsized product is 18 ounces. The counterfactual therefore never evaluates the cost function in the region where it is decreasing.

by abandoning downsizing, whether because attentive consumers make it less profitable or because a mandatory notice invites a consumer backlash. The first channel does not hold in our setting: McCormick’s variable profits under full attention with the downsized products exceed those under restored net weights by 10.5%, so downsizing stays profitable even when every consumer notices the reduction. Should the second channel operate, the outcome converges to that of a ban, and the gain from disclosure lies between the two figures we report. On consumer-surplus grounds, banning downsizing is therefore preferable only where disclosure is unavailable or ineffective.

Figure 8. : Welfare Change from Banning Downsizing



Notes: As in Figure 6a, each observation is a retailer–DMA–month containing at least one downsized product (18,628 markets); markets with no downsized product are unaffected by the ban and are excluded. Panel (a) reports the change in consumer surplus per consumer, in dollars, and panel (b) the same change as a percentage of per-capita expenditure on pepper in that market and month. Positive values indicate a gain. Marginal costs are adjusted using the quadratic specification of equation (11). Both histograms are trimmed at the 0.1st and 99.9th percentiles.

VIII. Conclusion

The practice of product downsizing occurs across a wide range of products and represents one strategy that firms use to increase unit prices. When consumers underuse size information or ignore unit prices, downsizing represents a hidden price increase. We utilize a downsizing event in the black pepper industry to determine whether consumers are inattentive to decreases in product size. The industry combines wide variation in package size with well-documented downsizing episodes, which together allow us to disentangle preferences for net weight from the degree to which consumers are inattentive.

To study how consumers respond to downsizing, we build a demand model that incorporates inattention to size changes and apply it to scanner data. In the model, inattentive consumers misperceive the net weights of the downsized products and as a result, they evaluate them based on their

original net weights. Because of this, the change in net weight enters utility as an additional product characteristic with a random coefficient, whose distribution we recover using standard demand estimation techniques.

We find that almost all consumers fail to notice the reduction in fill levels. Moreover, the estimated preferences suggest that even if consumers were fully aware, they would be more responsive to price than to product size. Inattention therefore makes downsizing more effective than it already is.

Inattention distorts product choices; its removal affects shares and consumer surplus. If consumers were fully attentive, the share of the downsized products would fall from 31 to 24 percent of the inside goods, a decline of 23%. Consumer surplus rises, on average, by 0.78% in the markets where downsizing occurred, and by more than 1 percent in over a quarter of them. Banning downsizing delivers only about half as much, 0.43%, because firms raise prices when they cannot reduce package sizes.

While packages display net weight, consumers misperceive how much they are buying: our estimates imply that almost none of them use this information at the point of sale. Mandatory labeling is therefore not sufficient on its own. This is why a policy that actively notifies consumers of a reduction, such as the French law requiring food retailers to flag downsized products (Rajbhandari and Adghirni, 2023), may benefit consumers. Our results further show that banning downsizing is the weaker instrument, and can be counterproductive. When unable to downsize, firms raise their package prices, an outcome consumers dislike more than smaller packages: restoring content costs roughly \$0.48 per ounce while consumers value an ounce at \$0.36. Under some cost specifications the price increase is large enough that a ban can even lower consumer surplus.

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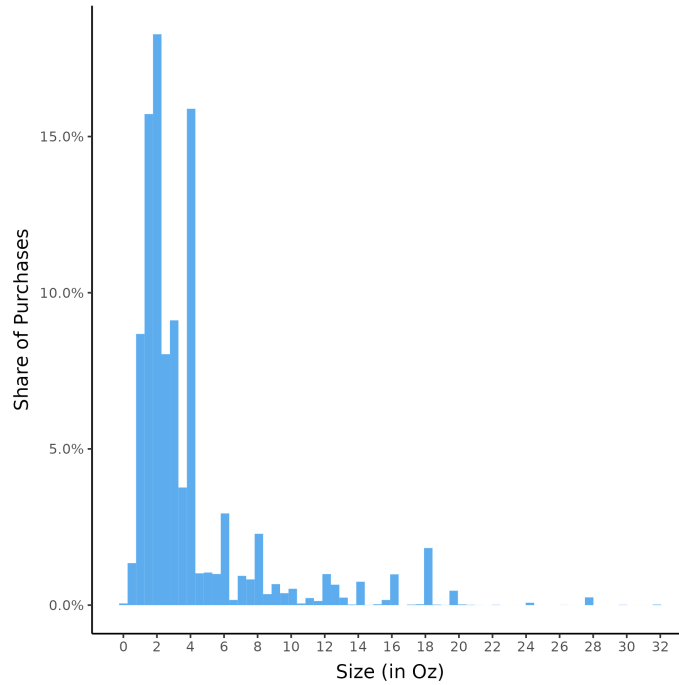
ADDITIONAL FIGURES AND TABLES

Table A1—: Pre-Downsizing Characteristics: Downsized vs. Non-Downsized McCormick Black Pepper

	Downsized	Non-downsized	<i>p</i> -value
Market share	0.0072	0.0025	0.007
Price (\$)	5.06	6.27	0.94
Package size (oz)	5.57	8.47	0.78
Whole (share)	0.40	0.33	0.76

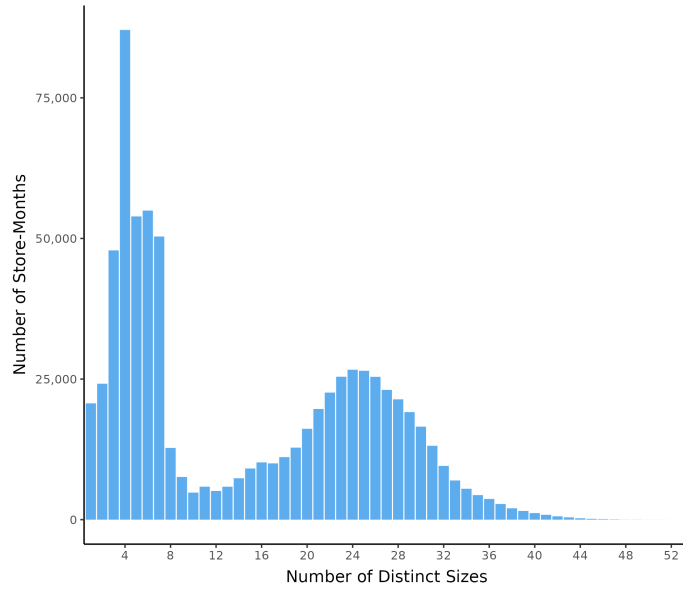
Mean pre-downsizing (pre-February 2015) characteristics of the downsized and non-downsized McCormick black-pepper products. *p*-values from Wilcoxon rank-sum (Mann–Whitney) tests.

Figure A1. : Distribution of Pepper Sizes Purchased by Households



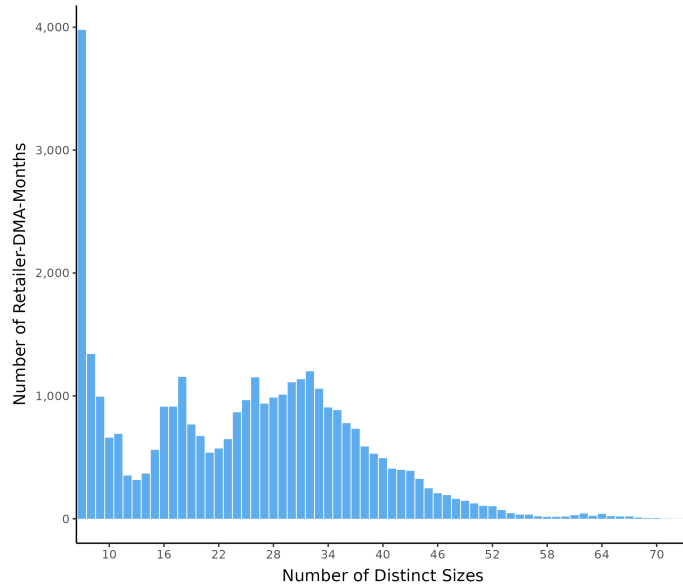
Distribution of net weight (in ounces) of pepper products purchased by households in the consumer panel, 2014–2016.

Figure A2. : Distribution of Distinct Sizes per Store-Month



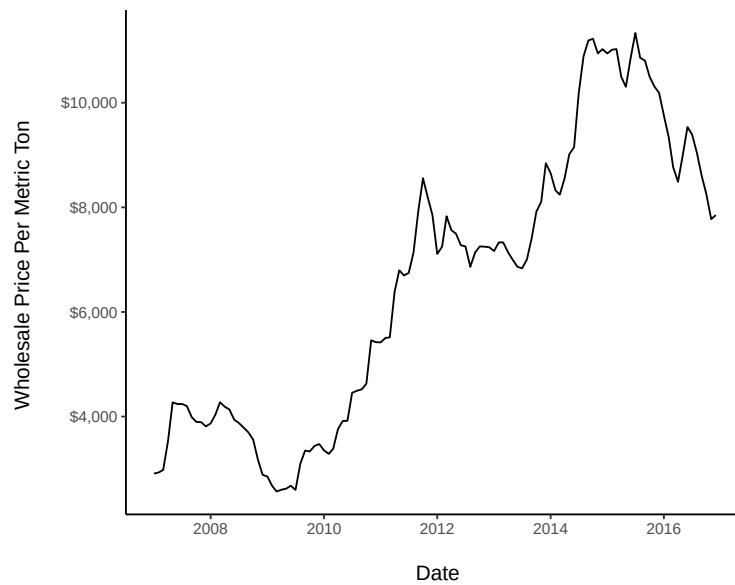
Distribution of the number of distinct pepper sizes offered per store-month across all stores in the scanner data, 2014–2016.

Figure A3. : Distribution of Distinct Sizes per Retailer-DMA-Month



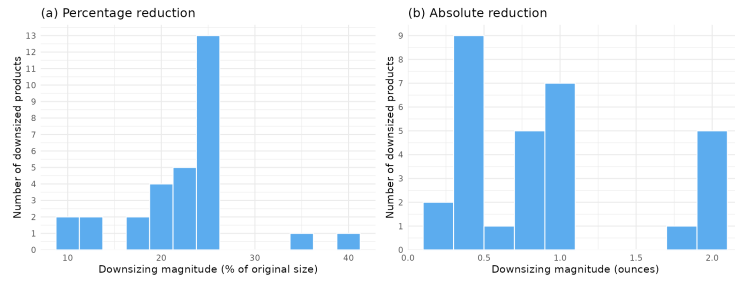
Distribution of the number of distinct pepper sizes offered per retailer-DMA-month in the estimation sample, 2014–2016. The minimum of 7 reflects the sample selection criterion.

Figure A4. : Wholesale Spot Price of Black Pepper in New York



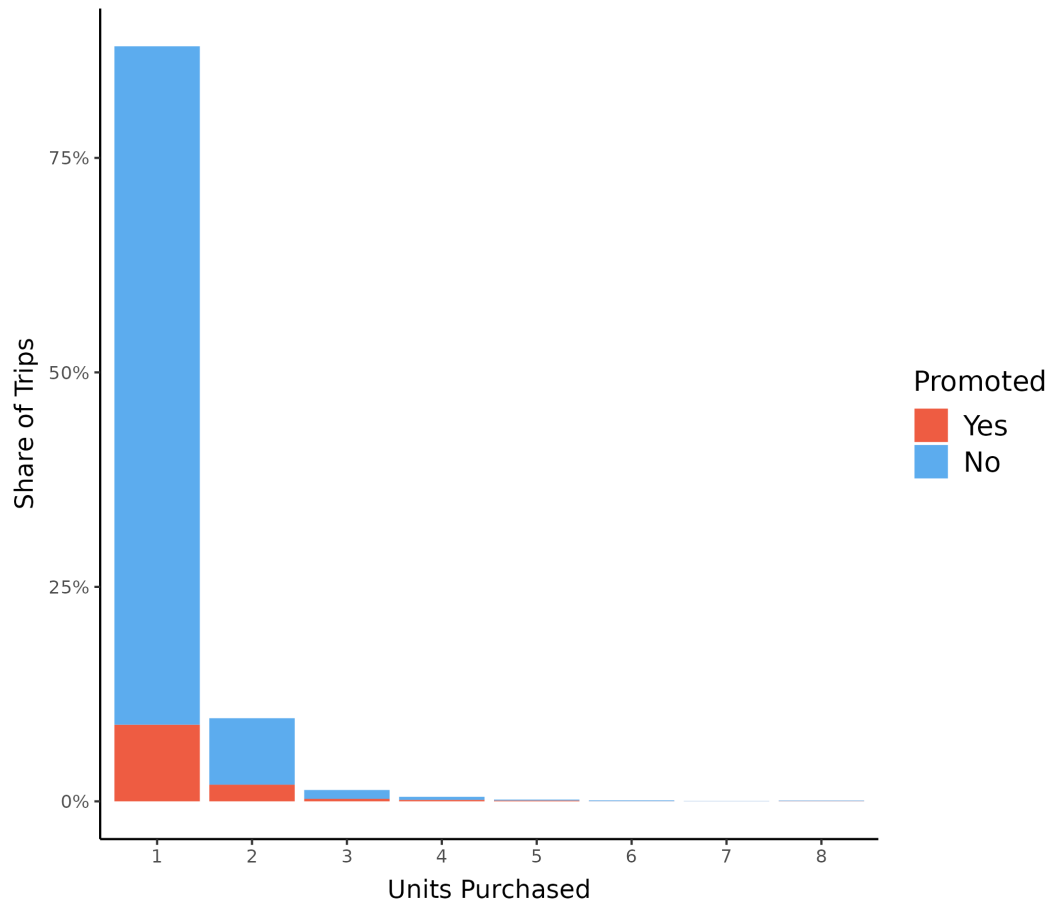
Monthly spot price of black pepper in New York. Source: Pepper Statistical Yearbook 2018, International Pepper Community.

Figure A5. : Distribution of Downsizing Magnitudes



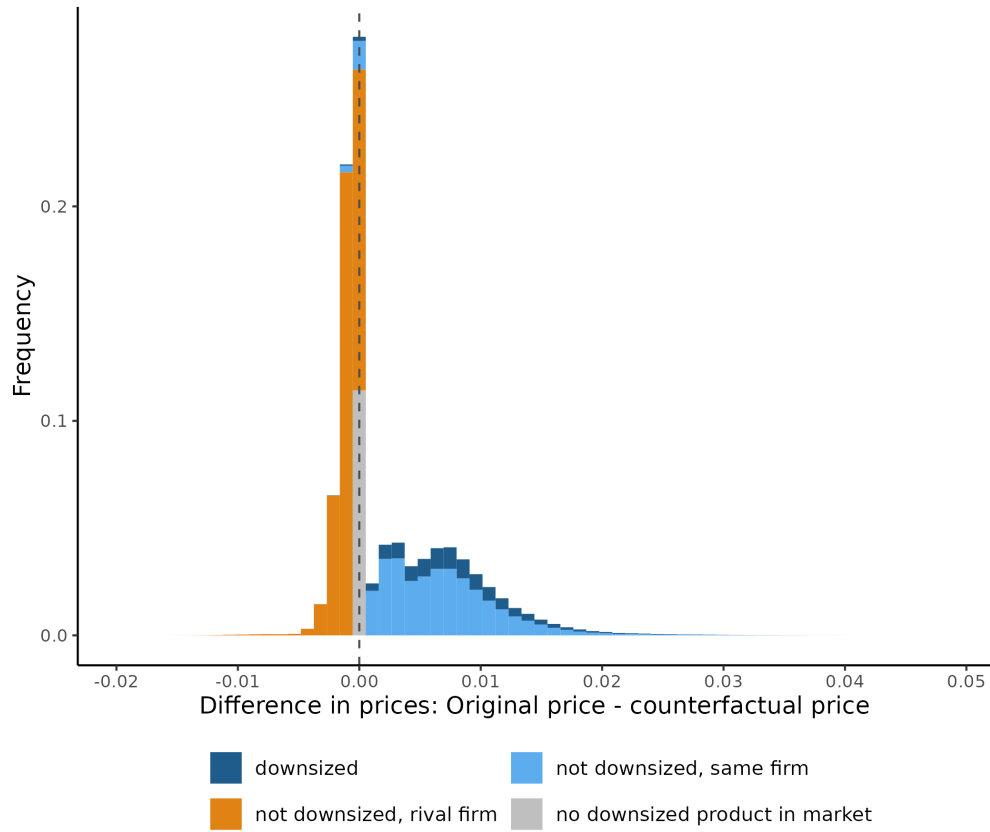
The distribution of downsizing magnitudes across the downsized products, measured (a) as a percentage of the original net weight and (b) in absolute ounces. Each bar counts the number of downsized products.

Figure A6. : Units Purchased per Shopping Trip



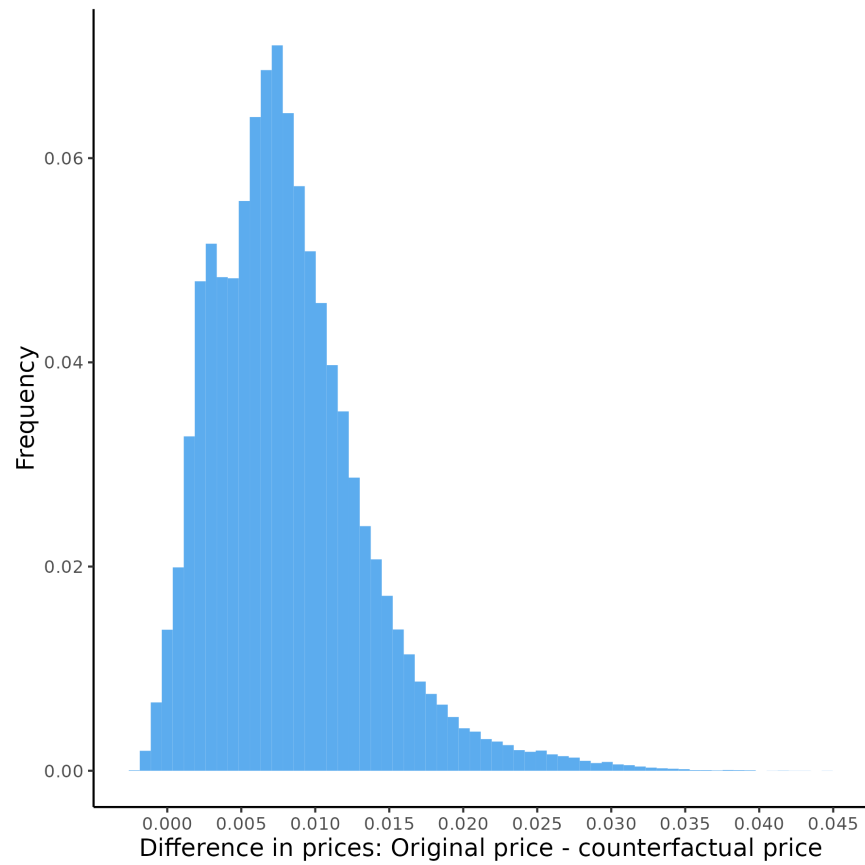
Notes: Distribution of the number of pepper units purchased per shopping trip, separately for promoted and non-promoted products. Only trips with a pepper purchase are included. Based on household panel data, 2014–2016.

Figure A7. : Counterfactual 1: Changes in Prices by Ownership



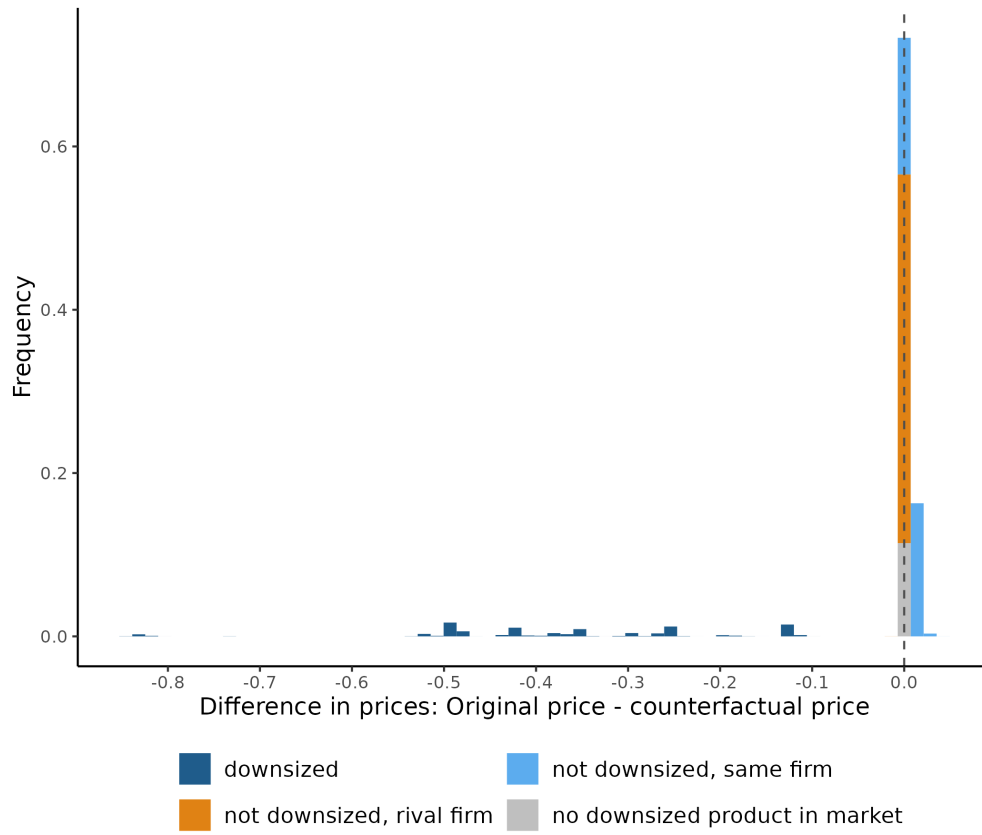
Notes: The horizontal axis is the observed price minus the counterfactual price, so positive values indicate that the price *falls* when inattention is removed. All products, 2015–2016. Products are grouped by ownership: those that were downsized, other products sold by the same firms, and products sold by firms that did not downsize. The mass at zero consists of markets stocking no downsized product together with rival products whose prices change by less than \$0.001.

Figure A8. : Counterfactual 1: Distribution of Changes in Prices (Downsized Products)



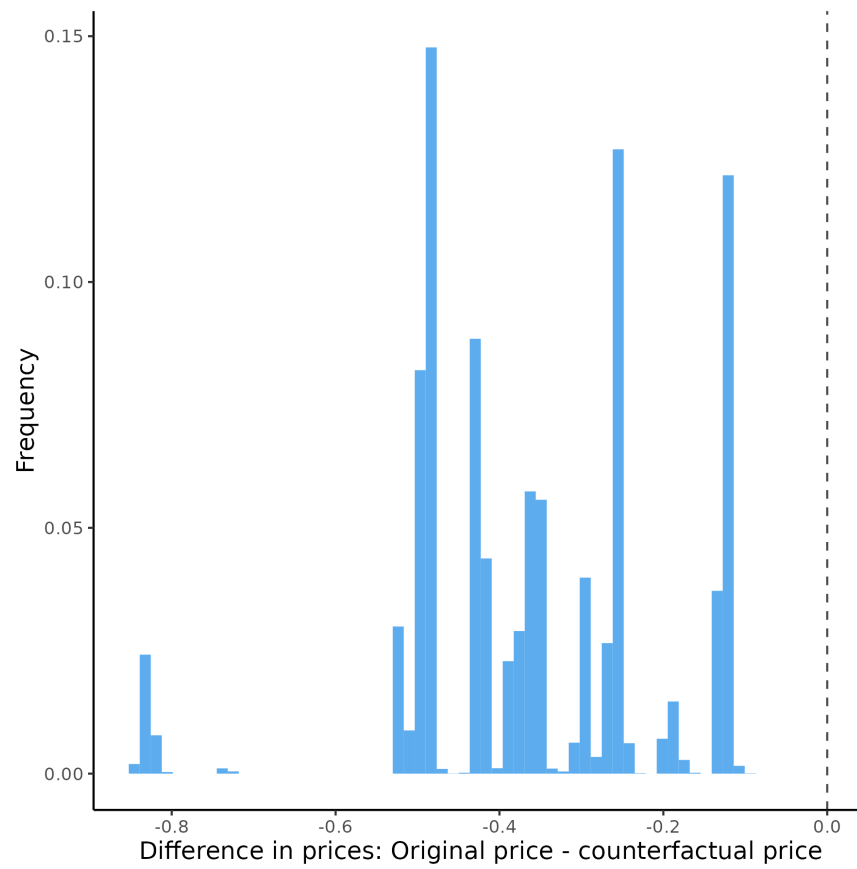
Notes: As in Figure A7, the horizontal axis is the observed price minus the counterfactual price, so positive values indicate that the price falls when inattention is removed. Restricted to the downsized products.

Figure A9. : Counterfactual 2: Changes in Prices by Ownership



Notes: As in Figure A7, the horizontal axis is the observed price minus the counterfactual price; the mass to the left of zero indicates that prices *rise* rather than fall. All products, 2015–2016, grouped by ownership as in Figure A7. Marginal costs are adjusted using the quadratic specification of equation (11).

Figure A10. : Counterfactual 2: Distribution of Changes in Prices (Downsized Products)



Notes: As in Figure A9, the horizontal axis is the observed price minus the counterfactual price, so negative values indicate that the price rises. Restricted to the products restored to their original net weight.

A1. Identification of Downsized Products

Downsized products typically receive new Universal Product Codes (UPCs), as the barcode encodes the package size. Because Nielsen tracks products at the UPC level, a size reduction that results in a new UPC creates a new product in the data rather than an update to an existing one. To identify which products in the data are downsized versions of existing products, we developed a procedure based on comparing unit sales patterns over time that consists of three steps. First, we restrict attention to products whose descriptions indicate they could be downsized versions of another product. Specifically, we look for pairs of UPCs that share the same brand, pepper type, and packaging format but differ in net weight. The downsized product must have a smaller net weight than the original.

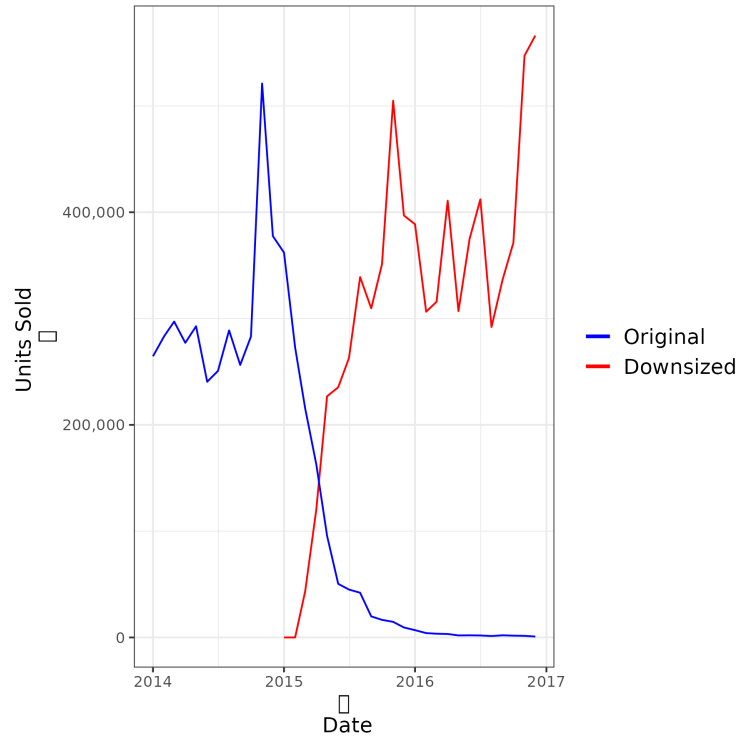
Second, we examine the time series of unit sales for each candidate pair. As retailers deplete their existing inventory of the original product and stock the new, smaller version, we expect to observe a characteristic pattern: declining sales of the original product accompanied by increasing sales of the slightly smaller product.

Third, for private-label products, we aggregate unit sales across stores within the same retailer before comparing time series, as individual store-level sales can be noisy. This aggregation is necessary because private-label UPCs are retailer-specific, and a given store may carry only one or two sizes within a brand, making product-level trends difficult to detect at the store level.

The court’s class-certification opinion lists the products at issue by UPC (In re: McCormick & Co., 2019), which we use to validate the downsized side of our candidate pairs. For the remaining pairs, we verified that the timing of the transition is consistent with known events. McCormick began shipping its downsized black pepper products in February 2015, in response to a spike in wholesale black pepper prices during 2014. Private-label brands followed shortly after: McCormick, which manufactures many private-label products, offered retailers the choice of accepting reduced weights or paying more to keep the original ones, and most accepted (In re: McCormick & Co., 2017).

Figure A11 illustrates the identification strategy for the McCormick medium tin of black pepper; the blue line shows the original 4-ounce product and the red line shows the downsized 3-ounce version. Sales of the original product decline sharply in early 2015 as the downsized version ramps up.

Figure A11. : Example of Sales Pattern: Original and Downsized Product



Monthly unit sales of a McCormick black pepper product. The blue line shows the original product (UPC 005210005030) and the red line shows the downsized version (UPC 005210002996).

From this procedure, we identified 30 episodes of downsizing: 15 name-brand and 15 private-label downsizing episodes. Table A2 provides a complete list of the name-brand downsized products and their size changes. All 15 belong to McCormick or its subsidiary brand (Spice Classics). Nearly all are black pepper, with the exception of one blend and one red pepper product. The size reductions range from 0.24 ounces (for a small grinder) to 2 ounces (for a large jar and a large tin), corresponding to percentage reductions between 11 and 35 percent. Spice Classics downsized two products, but the sales of these products are negligible; we therefore focus on downsizing by McCormick and private labels.

For name-brand products, the identification is straightforward because a given product shares the same UPC across all retailers. For private-label products, the same physical product can carry different UPCs at different retailers. We therefore aggregate unit sales across stores within the same retailer before comparing time series, as described in step three above.

Despite the small number of downsized products, these products collectively account for 34.1 percent of all pepper sales in the estimation sample. McCormick's downsized products account for 27.2 percent of sales and the private-label downsized products for 6.9 percent.

Table A2—: Downsized Products

	Product	UPC	Original (oz)	New (oz)	Decrease (oz)	Decrease (%)
1.	McCormick Large Tin	005210005040	8	6	2	25
2.	McCormick Large Jar	005210011656	18	16	2	11
3.	McCormick Large Jar	005210030441	18	16	2	11
4.	McCormick Medium Tin	005210005030	4	3	1	25
5.	McCormick Large Jar	005210009000	8.75	7.75	1	11
6.	Spice Classics Medium Jar	005210002073	2.5	1.62	0.88	35
7.	McCormick Medium Jar	005210007126	4	3.12	0.88	22
8.	McCormick Medium Jar	005210007128	4.25	3.5	0.75	18
9.	McCormick Medium Grinder	005210002357	3.1	2.5	0.6	19
10.	McCormick Medium Jar	005210000688	2.37	1.87	0.5	21
11.	McCormick Small Tin	005210005020	2	1.5	0.5	25
12.	McCormick Small Jar	005210000283	1.75	1.37	0.38	22
13.	Spice Classics Small Jar	005210000812	2	1.62	0.38	19
14.	McCormick Small Jar	005210000173	1.62	1.25	0.37	23
15.	McCormick Small Grinder	005210074603	1.24	1	0.24	19

Notes: Name-brand products only, ordered from largest to smallest absolute size change. All products are black pepper except row 12 (red) and row 14 (a blend).

ADDITIONAL ANALYSES

B1. Supporting Evidence on Time-Invariant Preferences

We split the pre-downsizing sample into January–June 2014 and July–December 2014. Columns (2) and (3) of Table B1 report the corresponding IV logit estimates. Panel B reports Wald tests of equality of the price and net-weight coefficients across the two sub-periods. We fail to reject equality for net weight ($p = 0.97$) and for price ($p = 0.47$).

We then compare pre-period estimates to those obtained after downsizing. Column (4) of Table B1 reports estimates for the full pre-period (before February 2015), column (5) for the overlap and post-downsizing period (February 2015 onward), and column (6) for the post period excluding the transition (after August 2015, when downsized products largely replace the original sizes).

In contrast to the within-pre-period results, we find sharp differences after downsizing. The net-weight coefficient declines by roughly 40%, from 0.457 in the pre-period to 0.274 in the overlap/post period and 0.207 in the post-only period, consistent with attenuation induced by inattention to downsizing. Panel B rejects equality of the pre- and post-period net weight coefficients ($p < 0.001$).

As an additional diagnostic, we estimate the logit model on the pre-downsizing sample allowing the coefficient on net weight to vary flexibly by month by interacting net weight with month dummies. Figure B1 plots the resulting month-by-net weight interaction coefficients. A joint Wald test fails to reject that these interactions are zero, consistent with a flat pre-trend in weight sensitivity.

Table B1—: Logit Demand Estimates Across Time Periods

Panel A: IV Logit Demand Estimates

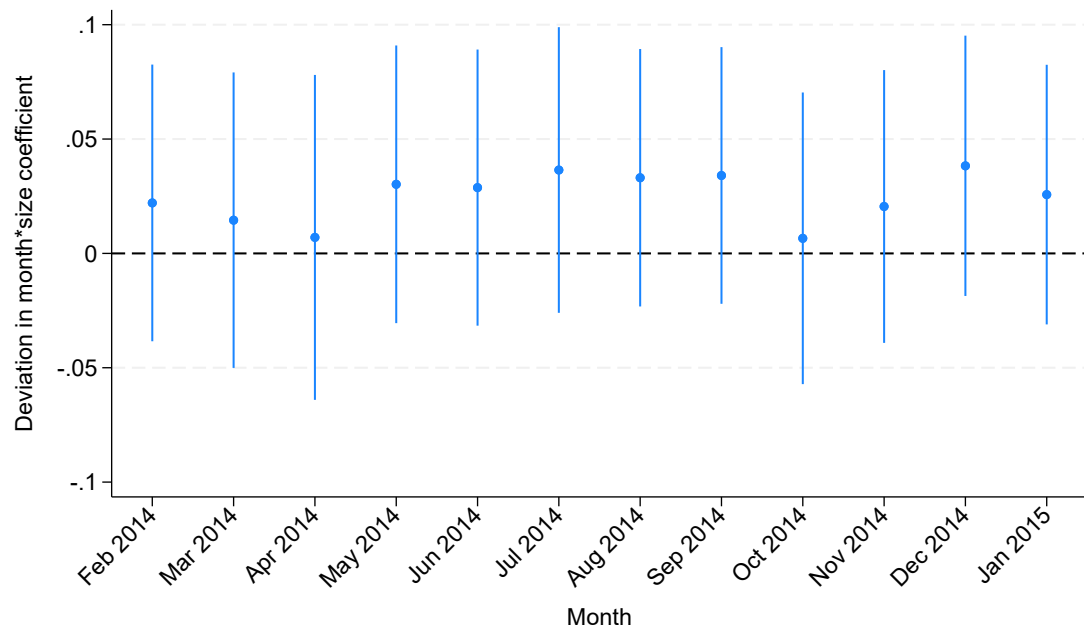
	(1) Full sample	(2) Pre first half	(3) Pre second half	(4) Pre full	(5) Overlap and post	(6) Post only
Price	−0.976*** (0.014)	−1.258*** (0.044)	−1.214*** (0.041)	−1.261*** (0.030)	−0.736*** (0.015)	−0.563*** (0.014)
Net weight	0.365*** (0.007)	0.447*** (0.023)	0.446*** (0.020)	0.457*** (0.015)	0.274*** (0.007)	0.207*** (0.007)
Observations	1,355,078	220,808	221,942	480,581	874,497	602,792

Panel B: Wald Tests of Coefficient Stability

Comparison	Net weight	Price
Pre, first half vs. pre, second half	0.002 [0.967]	0.520 [0.471]
Pre, full vs. overlap and post	117.933 [0.000]	245.904 [0.000]
Pre, full vs. post only	220.720 [0.000]	437.690 [0.000]

Notes: Dependent variable is $\ln(s_{jkt}/s_{0kt})$. Standard errors clustered at the market level in parentheses. Characteristic coefficients (constant, whole-peppercorn indicator, McCormick indicator and pepper-type indicators) are estimated but omitted. The pre-period runs to January 2015, split into January–June and July–December 2014; the overlap and post period begins in February 2015; the post-only period begins in September 2015. Panel B reports Wald F -statistics with p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure B1. : Pre-Period Stability of Net Weight Coefficient



Month-by-net-weight interaction coefficients from the IV logit regression estimated on the pre-downsizing period, relative to the omitted base month. Bars denote 95% confidence intervals.

B2. Monte Carlo Simulation

DATA GENERATING PROCESS

We simulate $T = 100$ markets, each having $J = 50$ products. Consumers evaluate product j in market t based on an exogenous characteristic $X_{jt} \sim N(0, 1)$ and an unobserved characteristic $\xi_{jt} \sim N(0, 0.3^2)$. We draw product sizes Z_{jt} from a uniform distribution between 2 and 16. Consumer i 's actual utility from consuming product j is:

$$(B1) \quad U_{ijt}^a = \beta_0 + X_{jt}\beta_1 + Z_{jt}\gamma - \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

where $(\beta_0, \beta_1, \gamma) = (-5, 1, 1)$, $\alpha_i \sim N(2, 0.5^2)$, and ε_{ijt} is distributed Type I extreme value.

To simulate downsizing, we randomly select 30% of the products and shrink the sizes of those products by 20%. We then randomly select which markets receive the downsized product. As in our data, some markets have the original products and some have the downsized ones.

We assume that the probability of being attentive is $\eta = 0.3$. As in our main model, inattention causes consumers to misjudge the size of a product. An inattentive consumer i *perceives* the utility of downsized product j based on its original size Z_{j0} as:

$$(B2) \quad U_{ijt}^p = \beta_0 + X_{jt}\beta_1 + Z_{j0}\gamma - \alpha_i p_{jt} + \xi_{jt} + \varepsilon_{ijt}$$

For the supply side, we assume that prices are set at marginal cost, as under perfect competition:

$$(B3) \quad p_{jt} = mc_{jt} = 0.5X_{jt} + Z_{jt} + W_{jt} + \xi_{jt} + \omega_{jt},$$

where $W_{jt} \sim N(0, 1)$ is an exogenous cost shifter and $\omega_{jt} \sim N(0, 0.1^2)$ a cost shock. Prices depend on the demand unobservable ξ_{jt} , so they are endogenous exactly as in the data, and W_{jt} serves as the instrument for price in the demand estimation. Marginal cost pricing is sufficient for our purposes because demand estimation does not use the supply model; what matters is that prices be correlated with ξ_{jt} , so that the simulation tests the estimator against the same endogeneity problem it faces in the data.

RESULTS

We estimate the model using the BLP procedure described in section V.B. The table below presents the results for 100 simulations. On average, our estimates are very close to the true values. Crucially, we accurately recover the attention parameter. The confidence interval for $\hat{\eta}$ is fairly narrow and covers the true value of $\eta = 0.3$.

B3. Inattention under Different Functional Form Assumptions

In this subsection, we derive the utility wedge due to inattention under alternative functional forms for net weight in utility.

As in the modeling section, M_{kt} consumers visit retailer k looking to buy pepper in period t . Each

Table B2—: Monte Carlo Results

	β_0	β_1	α	γ	σ_p	η
True value	−5	1	−2	1	0.5	0.3
Estimate	−4.95	1.00	−2.05	0.99	0.53	0.29
	[−5.36; −4.39]	[0.98; 1.02]	[−2.44; −1.68]	[0.98; 1.01]	[0.30; 0.75]	[0.24; 0.34]

Notes: Data generated as described in Section B.B2 with $T = 100$ markets and $J = 50$ products. Brackets report 95% confidence intervals across 100 replications.

consumer selects one product from the available pepper products J_{kt} or selects the no-purchase option 0. To simplify the exposition, we suppress the random coefficients; allowing them leaves each wedge unchanged with γ replaced by γ_i .

QUADRATIC SPECIFICATION

Suppose that net weight enters utility as a quadratic. Consumer i 's *actual* utility from purchasing product j is:

$$(B4) \quad U_{ijkt}^a = x_{jkt}\beta + \gamma_1 z_{jkt} + \gamma_2 z_{jkt}^2 - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt},$$

where x_{jkt} is a set of observable characteristics; p_{jkt} is the price; z_{jkt} is the current net weight, and z_{jk0} is the original (pre-downsizing) weight; ξ_{jkt} is an unobserved product attribute; and ε_{ijkt} is a random shock.

An *inattentive* consumer evaluates the downsized product j using its original weight and *perceives* their utility from j as:

$$\begin{aligned}
 (B5) \quad U_{ijkt}^p &= x_{jkt}\beta + \gamma_1 z_{jk0} + \gamma_2 z_{jk0}^2 - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} \\
 &= x_{jkt}\beta + \gamma_1 (z_{jkt} + z_{jk0} - z_{jkt}) + \gamma_2 (z_{jkt}^2 + z_{jk0}^2 - z_{jkt}^2) - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} \\
 &= x_{jkt}\beta + \gamma_1 z_{jkt} + \gamma_2 z_{jkt}^2 - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} + \gamma_1 \Delta_0 z_{jkt} + \gamma_2 \Delta_0 (z_{jkt}^2) \\
 &= U_{ijkt}^a + \gamma_1 \Delta_0 z_{jkt} + \gamma_2 \Delta_0 (z_{jkt}^2),
 \end{aligned}$$

where $\Delta_0 z_{jkt} = z_{jk0} - z_{jkt}$ is the change in the product weight and $\Delta_0 (z_{jkt}^2) = z_{jk0}^2 - z_{jkt}^2$ is the change in the squared product weight. The wedge term caused by inattention is therefore $\gamma_1 \Delta_0 z_{jkt} + \gamma_2 \Delta_0 (z_{jkt}^2)$. Because of inattention, the change in net weight and the change in the squared net weight enter perceived utility as additional characteristics.

LOGARITHMIC SPECIFICATION

Suppose instead that net weight enters utility logarithmically. Consumer i 's *actual* utility from purchasing product j is now:

$$(B6) \quad U_{ijkt}^a = x_{jkt}\beta + \gamma \log(z_{jkt}) - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt}$$

An *inattentive* consumer evaluates the downsized product j using its original weight and *perceives* their utility from j as:

$$\begin{aligned}
(B7) \quad U_{ijkt}^p &= x_{jkt}\beta + \gamma \log(z_{jk0}) - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} \\
&= x_{jkt}\beta + \gamma \log(z_{jk0}) + \gamma \log(z_{jkt}) - \gamma \log(z_{jkt}) - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} \\
&= x_{jkt}\beta + \gamma \log(z_{jkt}) - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} + \gamma \log\left(\frac{z_{jk0}}{z_{jkt}}\right) \\
&= U_{ijkt}^a + \gamma \log\left(\frac{z_{jk0}}{z_{jkt}}\right).
\end{aligned}$$

The utility wedge from inattention is therefore $\gamma \log\left(\frac{z_{jk0}}{z_{jkt}}\right)$.

WHOLE $\times$ SIZE INTERACTION

Suppose the marginal utility of net weight differs between whole and ground pepper. Consumer i 's *actual* utility from purchasing product j is now:

$$(B8) \quad U_{ijkt}^a = x_{jkt}\beta + \gamma z_{jkt} + \phi(w_j z_{jkt}) - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt}$$

where w_j is an indicator equal to one for whole peppercorns (a product characteristic that is stable over time) and ϕ is the interaction coefficient.

An *inattentive* consumer evaluates the downsized product j using its original weight in *both* the linear net-weight term and the interaction term (since w_j does not change), and *perceives* his utility from j as:

$$\begin{aligned}
(B9) \quad U_{ijkt}^p &= x_{jkt}\beta + \gamma z_{jk0} + \phi(w_j z_{jk0}) - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} \\
&= x_{jkt}\beta + \gamma z_{jk0} + \gamma z_{jkt} - \gamma z_{jkt} + \phi w_j z_{jk0} + \phi w_j z_{jkt} - \phi w_j z_{jkt} \\
&\quad - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} \\
&= x_{jkt}\beta + \gamma z_{jkt} + \phi w_j z_{jkt} - \alpha p_{jkt} + \xi_{jkt} + \varepsilon_{ijkt} + (\gamma + \phi w_j)(z_{jk0} - z_{jkt}) \\
&= U_{ijkt}^a + (\gamma + \phi w_j)(z_{jk0} - z_{jkt}).
\end{aligned}$$

The utility wedge from inattention is therefore $(\gamma + \phi w_j)(z_{jk0} - z_{jkt})$.

In sum, inattention generates a wedge of $\gamma \Delta_0 z_{jkt}$ for ground-pepper products and $(\gamma + \phi) \Delta_0 z_{jkt}$ for whole-pepper products.

B4. Marginal Cost Results

Table B3 reports the supply-side estimates. The first three columns report marginal cost in levels and the last three in logarithms; within each, net weight enters linearly, quadratically, or in logs.

Table B3—: Marginal Cost Results

	(1) mc	(2) mc	(3) mc	(4) log(mc)	(5) log(mc)	(6) log(mc)
Constant	3.103*** (0.377)	2.809*** (0.426)	2.943*** (0.446)	0.952*** (0.168)	0.770*** (0.206)	0.851*** (0.207)
Net Weight	0.384*** (0.025)	0.560*** (0.071)		0.102*** (0.008)	0.211*** (0.031)	
(Net Weight) ²		−0.010*** (0.004)			−0.006*** (0.001)	
log(Net Weight)			1.759*** (0.110)			0.574*** (0.069)
Is Whole	0.338 (0.259)	0.404 (0.251)	0.499** (0.235)	0.008 (0.098)	0.049 (0.097)	0.080 (0.096)
Black Pepper	−1.001*** (0.259)	−1.114*** (0.264)	−1.291*** (0.288)	−0.287*** (0.091)	−0.357*** (0.083)	−0.422*** (0.077)
Blend Pepper	−0.409 (0.543)	−0.502 (0.522)	−0.726 (0.541)	−0.022 (0.185)	−0.079 (0.174)	−0.147 (0.168)
Cayenne Pepper	−1.221*** (0.340)	−1.266*** (0.378)	−1.370*** (0.426)	−0.460*** (0.154)	−0.488*** (0.171)	−0.524*** (0.180)
Citrus Pepper	−2.009*** (0.297)	−2.079*** (0.288)	−2.302*** (0.274)	−0.657*** (0.167)	−0.700*** (0.154)	−0.803*** (0.132)
Garlic Pepper	−1.500*** (0.360)	−1.591*** (0.351)	−1.781*** (0.354)	−0.398** (0.155)	−0.454*** (0.146)	−0.517*** (0.138)
Red Pepper	−1.455*** (0.200)	−1.444*** (0.202)	−1.413*** (0.200)	−0.561*** (0.114)	−0.554*** (0.110)	−0.548*** (0.108)
Other Pepper	−0.485 (0.339)	−0.524 (0.334)	−0.525* (0.303)	−0.115 (0.148)	−0.139 (0.141)	−0.144 (0.127)
Observations	1,347,922	1,347,922	1,347,922	1,344,684	1,344,684	1,344,684
R^2	0.309	0.318	0.304	0.189	0.214	0.252
Implied Δmc (\$)	0.292	0.363	0.444	0.298	0.456	0.512

Notes: Standard errors clustered by manufacturer (345 clusters) in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The sample is restricted to package sizes between 0.4 and 32 ounces; columns (4)–(6) additionally exclude the 3,238 observations with non-positive recovered marginal cost. The final row reports the mean increase in marginal cost that each specification implies when downsized products are restored to their original net weight, averaged over the affected observations.

We cluster the standard errors at the manufacturer level; for private-label products, whose manufacturer we do not observe, we treat the retailer that sells them as the manufacturer. The coefficients

have the expected signs and marginal cost increases with net weight in every specification. Black, cayenne, citrus, garlic and red pepper are significantly cheaper than white pepper, which requires more processing. Blend and other varieties are also estimated to be cheaper, but imprecisely. The coefficient on whole spices is positive throughout but also imprecisely estimated.

Our main specification is the one reported in column (2). Linearity in net weight is rejected: $\kappa_3 = -0.0104$ with a clustered standard error of 0.0039 ($p = 0.008$). The estimated concavity is consistent with economies of scale in filling, packing, and distribution.

We run two further specifications. First, we nest the quadratic and the logarithmic forms in a common model, $mc = x\kappa_1 + z\kappa_2 + z^2\kappa_3 + \kappa_4 \log z + \omega$. The squared term becomes insignificant ($p = 0.65$) while $\log z$ retains significance ($p = 0.005$); the data favor a logarithmic curvature over a quadratic one. Second, we allow the dependent variable itself to be transformed. We use a Box-Cox transformation to rank the two specifications.²⁹ We obtain $\hat{\lambda} = 0.49$, between the two, so neither transformation is clearly preferred.³⁰

Neither check threatens the counterfactual, whose results depend not on which functional form fits best but on the cost increase each implies when packages are restored to their original weight. The bottom row of Table B3 reports that quantity. The two that are linear in net weight imply roughly \$0.29, but linearity is rejected in both the level and the logarithmic families (columns (2) and (5)); every specification admitting curvature implies at least \$0.36 in marginal cost increase. Our preferred specification yields the smallest of the latter, so the counterfactual in Section VII adopts a lower bound on the cost of a ban. The logarithmic specifications would imply a larger cost increase, and correspondingly a worse outcome for the policy.

²⁹The level and logarithmic specifications are not nested in one another, so no conventional test can rank them. The Box-Cox transformation embeds both in the family $(mc^\lambda - 1)/\lambda$, which reduces to marginal cost in levels when $\lambda = 1$ and to $\log(mc)$ as $\lambda \rightarrow 0$, and estimates λ by maximum likelihood rather than imposing it.

³⁰The transformation requires a positive dependent variable, so this estimate is computed on the sample used in columns (4)–(6), excluding observations with non-positive recovered marginal cost.